

Contents

2	Mathematical introduction	1
2.1	Introduction	1
2.2	A compact toolkit for jump processes	2
2.2.1	Exponential clocks and competing events	2
2.2.2	Continuous-time Markov chains and generators	3
2.2.3	Probability generating functions	4
2.3	The Galton–Watson process	4
2.3.1	Binary reproduction	5
2.3.2	Critical decay and extinction time	7
2.4	Conditioning on survival and quasi-stationarity	9
2.4.1	The late survival scale	9
2.4.2	The limiting conditional mean	10
2.4.3	The limiting conditional variance	10
2.4.4	A continuum approximation to the recursion	11
2.5	Small founding populations and competing jumps	11
2.5.1	Early survival and k founders	12
2.5.2	A discrete rupture mechanism	12
2.5.3	The linear continuous-time birth–death process	14
2.5.4	Birth–death with catastrophe as a killed chain	16
2.6	The survival amplitude $A(p)$	17
2.6.1	An infinite product	17
2.6.2	An exact series and two-sided bounds	18
2.6.3	The near-critical limit	20
2.6.4	Discrete and continuous time	21
2.6.5	Koenigs linearisation	21
2.6.6	Computation in practice	23
2.7	Absorption models and the method of characteristics	24
2.7.1	Transfer only	24
2.7.2	Transfer followed by death	25
2.7.3	Characteristics tested on the transfer–death PDE	27
2.7.4	Transfer, birth and death	27
2.7.5	Recovering state probabilities	31
	Chapter-local appendices	32
2.A	Closed-form searches and quadratic conjugacy	32

2.A.1	Exploratory formulae	32
2.A.2	Conjugacy with the quadratic family	33
2.A.3	The two elementary comparison cases	33
2.B	An exponential–rational hypergeometric integral	35
2.C	Coefficient extraction from the generating functions	36
2.C.1	A multilinear power	37
2.C.2	Transfer, birth and death	38

Chapter 2

Mathematical introduction

2.1 Introduction

When a population contains only a few individuals, its extinction, persistence and time spent near zero are distributional questions rather than small corrections to a deterministic mean. The models used later in this thesis therefore describe integer counts that change at random event times. Birth, death, transfer, absorption and rupture are treated as abstract jump mechanisms, while the population count records the state on which those mechanisms act.

This chapter develops two connected strands of the required probability. The first begins in discrete time with a binary Galton–Watson process. Its simplicity makes the distinction between mean behaviour and survival particularly sharp. Below the critical reproduction probability the population becomes extinct almost surely, but the population conditional on survival approaches a non-degenerate limiting law. For the binary model, the limiting conditional mean is the reciprocal of the constant

$$A(p) = \lim_{n \rightarrow \infty} \frac{S_n}{(2p)^n}, \quad 0 < p < \frac{1}{2}, \quad (2.1.1)$$

where S_n is the probability of survival to generation n . The value at the left endpoint will always mean the right limit, $A(0) := \lim_{p \downarrow 0} A(p)$. The constant in (2.1.1) measures the cumulative effect of the nonlinear part of the survival recursion before its late geometric decay becomes dominant. It has an exact product representation, an exact series, useful two-sided bounds and a controlled near-critical asymptotic. It also has a less obvious interpretation as a value of a Koenigs linearising function. These facts explain both how $A(p)$ can be computed and why elementary curve fitting has proved unreliable.

The second strand concerns continuous-time jump processes and their generating functions. Once a process has more than one count, its forward equations form a large or infinite coupled system. A multivariate probability generating function compresses that system into a first-order partial differential equation. The method of characteristics then converts the PDE into ordinary differential equations along curves in the enlarged state–time space. We develop this method through three transfer models. The first conserves particles, the second permits death after transfer, and the third adds linear birth. The final model has an infinite state space and leads to a closed generating function involving a Gauss hypergeometric function. The calculation supplies the analytical template used later for birth–death–catastrophe processes.

The two strands meet in several places. Both use generating functions to replace a countable family of probabilities by one analytic object. Both are organised by absorption: extinction in the branching process and transfer, death or rupture in the continuous-time models. Most importantly, both distinguish an unconditional population, which may decay to zero, from a population observed under a survival or non-absorption condition. Quasi-stationarity is the language for that distinction.

The order of the chapter follows the mathematical dependence rather than the later applications. [Section 2.2](#) gives only the continuous-time and generating-function background needed for the calculations that follow. The Galton–Watson process, its extinction threshold and its critical lifetime are developed in [section 2.3](#). [Section 2.4](#) introduces quasi-stationarity and obtains the limiting conditional moments. Founding populations, a discrete rupture mechanism, the linear continuous-time birth–death process and the killed-generator form of a catastrophe mechanism are treated in [section 2.5](#). The analysis of $A(p)$ occupies [section 2.6](#); the main text contains the exact results and proofs, while the exploratory closed-form search is retained in a chapter-local appendix. Finally, [section 2.7](#) derives and solves the generating-function PDEs. Two further appendices collect a hypergeometric integral identity and the extraction of state probabilities from multivariate generating functions.

Standard background is stated concisely. Proofs are given when the result is specific to the models developed here, and numerical evidence is labelled as such. This separation matters near criticality, where a visually accurate approximation to $A(p)$ can still produce a poor approximation to the conditional mean $1/A(p)$.

2.2 A compact toolkit for jump processes

Only a small amount of general machinery is needed before turning to the models. The essential objects are exponential clocks, continuous-time Markov chains, generators and probability generating functions. This section fixes the notation and, in particular, the convention used for forward equations.

2.2.1 Exponential clocks and competing events

A non-negative random variable T has an exponential distribution with rate $\lambda > 0$, written $T \sim \text{Exp}(\lambda)$, when

$$\mathbb{P}(T > t) = e^{-\lambda t}, \quad f_T(t) = \lambda e^{-\lambda t}, \quad t \geq 0. \quad (2.2.1)$$

The mean waiting time is $1/\lambda$. The defining survival function also gives the memoryless property

$$\mathbb{P}(T > s + t \mid T > s) = \mathbb{P}(T > t), \quad s, t \geq 0. \quad (2.2.2)$$

Consequently, after a period during which no event has occurred, the residual waiting time has the same law as it had initially.

The other fact used repeatedly is the exponential race. Let $T_1, \dots, T_m$ be independent exponential random variables with respective rates $a_1, \dots, a_m$, and write $T = \min_r T_r$. Independence gives

$$\mathbb{P}(T > t) = \prod_{r=1}^m \mathbb{P}(T_r > t) = \exp\left(-t \sum_{r=1}^m a_r\right), \quad (2.2.3)$$

so that $T \sim \text{Exp}(a_0)$, where $a_0 = \sum_r a_r$. Moreover,

$$\mathbb{P}(T_r = T) = \int_0^\infty a_r e^{-a_r t} \prod_{s \neq r} e^{-a_s t} dt = \frac{a_r}{a_0}. \quad (2.2.4)$$

Thus the total rate determines the waiting time and the relative rates determine which event occurs. State-dependent versions of these two statements provide a pathwise construction of the jump processes below.

2.2.2 Continuous-time Markov chains and generators

Let $X = (X_t)_{t \geq 0}$ take values in a countable state space E . It is a time-homogeneous continuous-time Markov chain if, for $s, t \geq 0$, its future conditional on the past depends only on the present state and the transition probabilities depend only on the elapsed time:

$$\mathbb{P}(X_{s+t} = j \mid X_s = i, X_u, 0 \leq u < s) = \mathbb{P}(X_t = j \mid X_0 = i) =: P_{ij}(t). \quad (2.2.5)$$

The transition matrices $P(t) = (P_{ij}(t))$ form a semigroup, $P(s+t) = P(s)P(t)$, with $P(0) = I$.

The generator $Q = (q_{ij})$ records the short-time transition rates. With the convention that q_{ij} is the rate from i to j ,

$$q_{ij} = \lim_{h \downarrow 0} \frac{P_{ij}(h) - \delta_{ij}}{h} \quad (i \neq j), \quad q_{ii} = -\sum_{j \neq i} q_{ij}. \quad (2.2.6)$$

Hence $P_{ij}(h) = q_{ij}h + o(h)$ for $j \neq i$, while $P_{ii}(h) = 1 + q_{ii}h + o(h)$. If the chain is in state i , it waits an exponential time with rate $q_i = -q_{ii} = \sum_{j \neq i} q_{ij}$ and then moves to $j \neq i$ with probability q_{ij}/q_i . A state with $q_i = 0$ is absorbing.

For a test function $f : E \rightarrow \mathbb{R}$, the generator acts as

$$(\mathcal{L}f)(i) = \sum_{j \neq i} q_{ij} \{f(j) - f(i)\}. \quad (2.2.7)$$

This form makes the event structure visible: each possible jump contributes its rate multiplied by the corresponding increment of f .

Write $p_j(t) = \mathbb{P}(X_t = j)$ and regard $\mathbf{p}(t) = (p_j(t))_{j \in E}$ as a row vector. The forward equations are

$$\frac{dp_j}{dt} = \sum_{i \neq j} p_i(t) q_{ij} - p_j(t) \sum_{k \neq j} q_{jk}, \quad \mathbf{p}'(t) = \mathbf{p}(t) Q. \quad (2.2.8)$$

The first sum is probability flow into state j ; the second is flow out. For the transition matrix itself, the backward and forward equations are respectively

$$P'(t) = QP(t), \quad P'(t) = P(t)Q. \quad (2.2.9)$$

Both hold for the time-homogeneous semigroup under the usual non-explosion and domain conditions. We will work almost entirely with the componentwise forward equations (2.2.8), where the convention is unambiguous.

2.2.3 Probability generating functions

If Z is a random variable on $\mathbb{Z}_{\geq 0}$, its probability generating function is

$$G_Z(z) = \mathbb{E}[z^Z] = \sum_{n=0}^{\infty} \mathbb{P}(Z = n)z^n, \quad |z| \leq 1. \quad (2.2.10)$$

The distribution and its first two moments can be recovered from

$$\mathbb{P}(Z = n) = \frac{G_Z^{(n)}(0)}{n!}, \quad \mathbb{E}[Z] = G'_Z(1), \quad \text{Var}(Z) = G''_Z(1) + G'_Z(1) - G'_Z(1)^2. \quad (2.2.11)$$

In particular, $G_Z(0) = \mathbb{P}(Z = 0)$. If $Z_1, \dots, Z_k$ are independent, then the generating function of their sum is the product of their generating functions. This elementary factorisation is the analytical form of the branching property used in [section 2.3](#).

For a process $X_t \in \mathbb{Z}_{\geq 0}$, the time-dependent generating function

$$G(z, t) = \mathbb{E}[z^{X_t}] = \sum_{n \geq 0} p_n(t)z^n \quad (2.2.12)$$

often converts the infinite system (2.2.8) into one PDE. For example, if state n jumps to $n + 1$ at rate $n\lambda$ and to $n - 1$ at rate $n\mu$, multiplication of the forward equations by z^n and summation give

$$\frac{\partial G}{\partial t} = (\lambda z - \mu)(z - 1) \frac{\partial G}{\partial z}. \quad (2.2.13)$$

The same operation in two dimensions uses

$$G(x, y, t) = \mathbb{E}[x^{X_t} y^{Y_t}] = \sum_{i, j \geq 0} p_{i, j}(t) x^i y^j. \quad (2.2.14)$$

Partial derivatives then encode factors of i and j . This is why linear per-capita event rates lead to first-order PDEs, and why the method of characteristics is the natural solution method in [section 2.7](#).

The generating-function manipulations below may first be read as operations on formal power series. Whenever analytic evaluation is used, the arguments are kept inside the unit disc or an analytic continuation is stated explicitly. This distinction becomes important for the hypergeometric representation in [section 2.7.4.3](#).

2.3 The Galton–Watson process

The Galton–Watson process is the simplest setting in which independent reproduction, extinction and conditioning can be seen together. Let $\xi, \xi_{n,i}$ be independent and identically distributed random variables with

$$\mathbb{P}(\xi = k) = p_k, \quad k \in \mathbb{Z}_{\geq 0}, \quad m = \mathbb{E}[\xi] < \infty. \quad (2.3.1)$$

Starting from one founder, $Z_0 = 1$, define

$$Z_{n+1} = \sum_{i=1}^{Z_n} \xi_{n,i}, \quad (2.3.2)$$

where the empty sum is zero. State 0 is absorbing. Conditional expectation in (2.3.2) gives

$$\mathbb{E}[Z_{n+1} \mid Z_n] = mZ_n, \quad \mathbb{E}[Z_n] = m^n. \quad (2.3.3)$$

Let

$$\phi(z) = \mathbb{E}[z^\xi] = \sum_{k \geq 0} p_k z^k \quad (2.3.4)$$

be the offspring generating function. The branching property implies that the generating function of Z_n is the n -fold composition $\phi_n = \phi \circ \cdots \circ \phi$. In particular, if

$$u_n = \mathbb{P}(Z_n = 0), \quad (2.3.5)$$

then

$$u_{n+1} = \phi(u_n), \quad u_0 = 0. \quad (2.3.6)$$

The limiting extinction probability is the smallest fixed point of ϕ in $[0, 1]$; see, for example, [1, 2].

2.3.1 Binary reproduction

The remainder of the discrete-time analysis uses the binary offspring law

$$\mathbb{P}(\xi = 2) = p, \quad \mathbb{P}(\xi = 0) = 1 - p, \quad 0 \leq p \leq 1. \quad (2.3.7)$$

Thus

$$\phi(z) = 1 - p + pz^2, \quad m = 2p, \quad \mathbb{E}[Z_n] = (2p)^n. \quad (2.3.8)$$

The process is subcritical, critical or supercritical according as $2p$ is less than, equal to or greater than one. Typical genealogies on either side of the threshold are shown in figure 2.1.

For the binary law, (2.3.6) becomes

$$u_{n+1} = 1 - p + pu_n^2. \quad (2.3.9)$$

It is convenient to use the survival probability

$$S_n = 1 - u_n = \mathbb{P}(Z_n > 0). \quad (2.3.10)$$

Substitution into (2.3.9) gives the logistic recursion

$$S_{n+1} = 2pS_n - pS_n^2, \quad S_0 = 1. \quad (2.3.11)$$

The sequence u_n is non-decreasing because extinction is absorbing; hence S_n is non-increasing and has a limit. Any such limit is a fixed point of (2.3.11). Solving

$$S = 2pS - pS^2 \quad (2.3.12)$$

gives

$$S = 0 \quad \text{or} \quad S = \frac{2p - 1}{p}. \quad (2.3.13)$$

The second root is inadmissible for $p < 1/2$, coalesces with zero at $p = 1/2$, and lies in $(0, 1]$ for $p > 1/2$. It follows that

$$S_\infty := \mathbb{P}(Z_n > 0 \text{ for every } n) = \begin{cases} 0, & 0 \leq p \leq \frac{1}{2}, \\ \frac{2p - 1}{p}, & \frac{1}{2} < p \leq 1. \end{cases} \quad (2.3.14)$$

The phase transition is continuous, but the stability of the zero fixed point changes at $p = 1/2$, as figure 2.2 makes explicit.

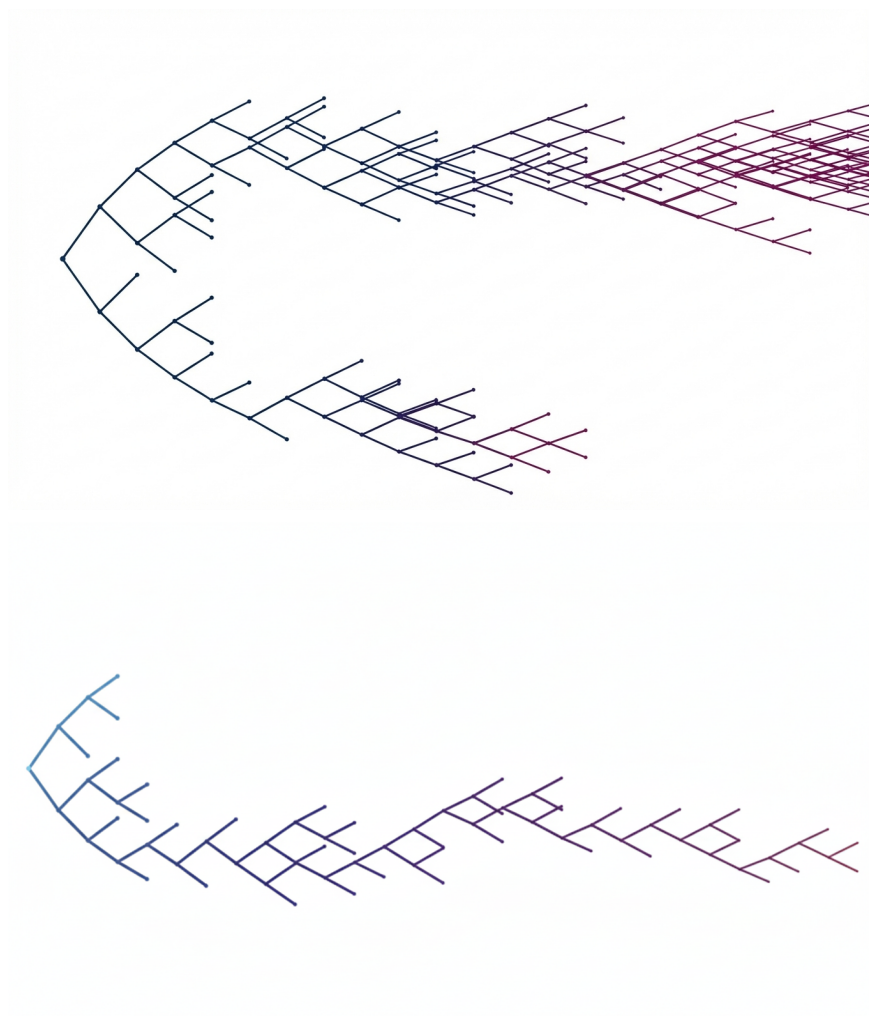


Figure 2.1: Realisations of the binary Galton–Watson process, with generation increasing from left to right. The upper realisation is supercritical ($p = 0.58$); the lower is subcritical ($p = 0.47$). Criticality concerns the law of the whole tree, not the appearance of any one finite realisation.

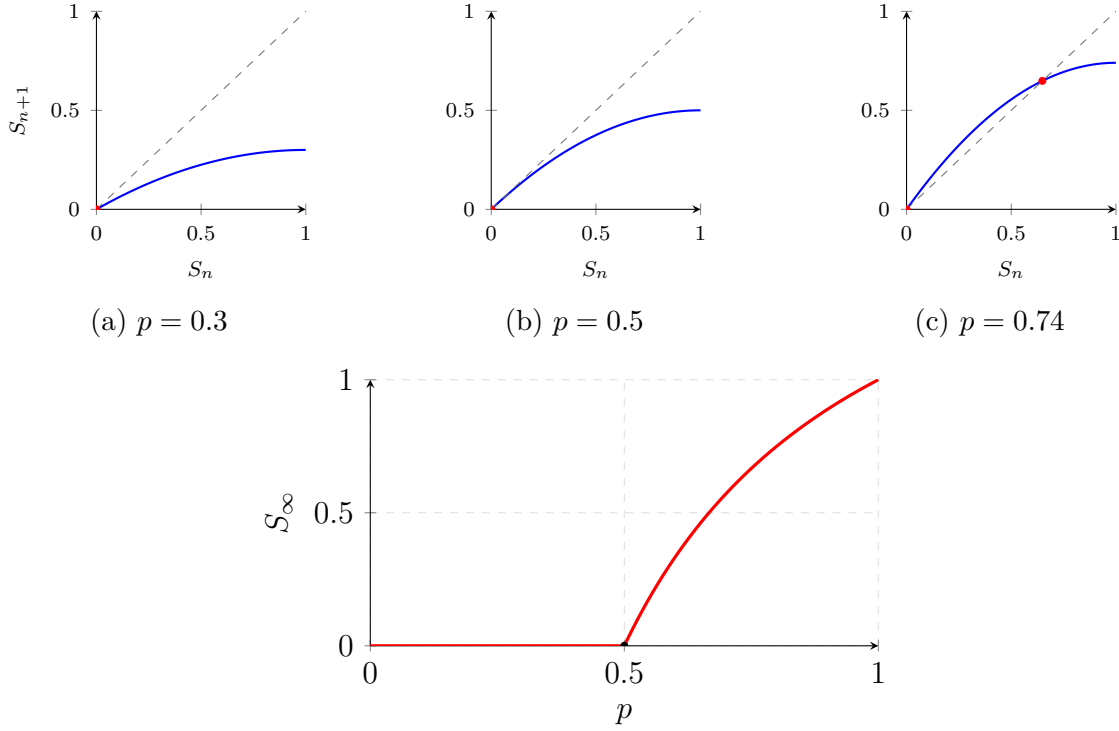


Figure 2.2: The survival map $S \mapsto 2pS - pS^2$ against the diagonal and the resulting ultimate survival probability. At criticality the map is tangent to the diagonal at the origin.

2.3.2 Critical decay and extinction time

At $p = 1/2$, the mean in (2.3.8) is one at every generation, even though (2.3.14) says that extinction is certain. The resolution is a heavy tail: rare surviving trees are sufficiently large to keep the unconditional mean fixed. The corresponding survival asymptotic can be obtained directly from the exact recursion, without a continuum approximation.

Proposition 2.1 (Critical survival probability). *For the critical binary process,*

$$S_n \sim \frac{2}{n} \quad (n \rightarrow \infty). \quad (2.3.15)$$

Proof. At $p = 1/2$, (2.3.11) reads

$$S_{n+1} = S_n - \frac{1}{2}S_n^2 = S_n \left(1 - \frac{S_n}{2}\right). \quad (2.3.16)$$

Because $S_n \downarrow 0$, the reciprocal sequence tends to infinity. Its increments satisfy

$$\frac{1}{S_{n+1}} - \frac{1}{S_n} = \frac{1}{2(1 - S_n/2)} \rightarrow \frac{1}{2}. \quad (2.3.17)$$

The Stolz–Cesàro theorem, or summation of the convergent increments, now gives $(1/S_n)/n \rightarrow 1/2$, which is equivalent to (2.3.15). $\square$

Let

$$T = \inf\{n \geq 0 : Z_n = 0\} \quad (2.3.18)$$

be the extinction time. Since $T = \sum_{n \geq 0} \mathbf{1}_{\{T > n\}}$, Tonelli's theorem gives the tail-sum identity

$$\mathbb{E}[T] = \sum_{n=0}^{\infty} \mathbb{P}(T > n) = \sum_{n=0}^{\infty} S_n. \quad (2.3.19)$$

By (2.3.15), the final series has the same divergent tail as $2 \sum_n n^{-1}$. Thus extinction occurs almost surely but

$$\mathbb{E}[T] = \infty. \quad (2.3.20)$$

The point probabilities follow from the exact decrement in (2.3.16):

$$\mathbb{P}(T = n + 1) = S_n - S_{n+1} = \frac{1}{2} S_n^2 \sim \frac{2}{n^2}. \quad (2.3.21)$$

The total progeny

$$\mathcal{T} = \sum_{n \geq 0} Z_n \quad (2.3.22)$$

has a different critical exponent. A finite binary genealogy with k division vertices contains $k + 1$ terminal vertices and hence $2k + 1$ individuals in all. The number of such ordered full binary trees is the Catalan number $C_k = (k + 1)^{-1} \binom{2k}{k}$. Consequently,

$$\mathbb{P}(\mathcal{T} = 2k + 1) = C_k p^k (1 - p)^{k+1}. \quad (2.3.23)$$

At $p = 1/2$, Stirling's formula yields

$$\mathbb{P}(\mathcal{T} = 2k + 1) \sim \frac{1}{2\sqrt{\pi}} k^{-3/2}. \quad (2.3.24)$$

The $3/2$ law therefore holds along the admissible odd values; the probability is zero at even values. Figure 2.3 places the survival, extinction-time and total-progeny scalings side by side.

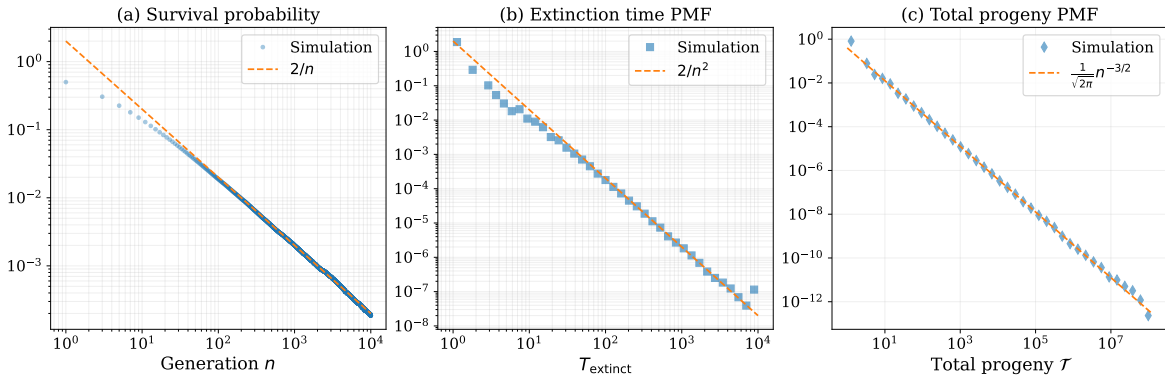


Figure 2.3: Critical power laws from 10^6 simulated binary Galton–Watson realisations. From left to right: $S_n \sim 2/n$, $\mathbb{P}(T = n) \sim 2/n^2$, and the $n^{-3/2}$ total-progeny law on its odd support. The first two exponents also follow directly from (2.3.16).

2.4 Conditioning on survival and quasi-stationarity

Almost-sure extinction does not imply that every surviving realisation is small. For a subcritical branching process the unconditional law concentrates at zero, whereas the law conditional on $Z_n > 0$ may approach a stable non-degenerate shape. That conditional limit is the object required here.

Definition 2.2 (Quasi-stationary and Yaglom laws). Let X_n be a Markov chain with absorbing state 0. A probability distribution ν on the non-absorbing states is *quasi-stationary* if

$$\mathbb{P}_\nu(X_n = j \mid X_n \neq 0) = \nu_j \quad (n \geq 0, j \neq 0). \quad (2.4.1)$$

If, from a specified initial state,

$$\mathbb{P}(X_n = j \mid X_n \neq 0) \longrightarrow \nu_j, \quad (2.4.2)$$

then ν is the corresponding Yaglom limit.

For a subcritical Galton–Watson process satisfying the usual non-degeneracy and integrability conditions, the Yaglom limit exists; the original result is due to Yaglom [3], with modern treatments in [1, 2]. The binary offspring law satisfies those conditions. We do not re-prove the general convergence theorem, but we derive the survival scale and the first two limiting moments explicitly. This distinction is important: convergence of two moments is consistent with the Yaglom theorem, but does not by itself prove convergence of the full conditional distribution.

2.4.1 The late survival scale

Fix $0 < p < 1/2$ and put $m = 2p \in (0, 1)$. The survival recursion can be written

$$S_{n+1} = mS_n \left(1 - \frac{S_n}{2}\right). \quad (2.4.3)$$

Since $S_n \downarrow 0$, the ratio S_{n+1}/S_n tends to m . The geometric rate is therefore m^n , but the ratio limit alone does not identify its amplitude. Define

$$A(p) = \lim_{n \rightarrow \infty} \frac{S_n}{m^n}. \quad (2.4.4)$$

The product argument in [section 2.6.1](#) proves that this limit exists and lies strictly between zero and one half. Consequently,

$$S_n \sim A(p)m^n \quad (n \rightarrow \infty). \quad (2.4.5)$$

The relation is an asymptotic equivalence: the ratio of its two sides tends to one. It is not a formula for the limit of S_n , which is zero.

The constant $A(p)$ retains the effect of the early nonlinear iterations in (2.4.3). Once S_n is small, the factor $1 - S_n/2$ is close to one and subsequent decay is nearly linear. The accumulation of all earlier factors is precisely what remains in $A(p)$.

2.4.2 The limiting conditional mean

Because $Z_n = 0$ contributes nothing to its own expectation,

$$\mathbb{E}[Z_n] = \mathbb{P}(Z_n > 0)\mathbb{E}[Z_n \mid Z_n > 0]. \quad (2.4.6)$$

Using (2.3.8) and (2.3.10) gives the exact identity

$$\mathbb{E}[Z_n \mid Z_n > 0] = \frac{m^n}{S_n}. \quad (2.4.7)$$

The asymptotic (2.4.5) therefore yields

$$\mathbb{E}[Z_n \mid Z_n > 0] \longrightarrow \frac{1}{A(p)}. \quad (2.4.8)$$

The geometric decay of the unconditional mean is exactly cancelled by that of the survival probability. The surviving population consequently has a finite limiting mean even though the population as a whole becomes extinct almost surely.

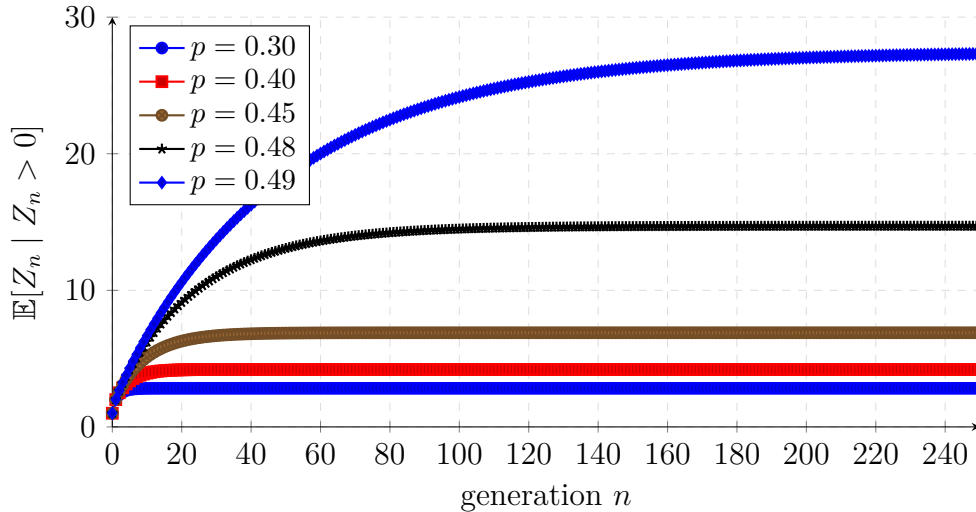


Figure 2.4: Conditional means computed from the exact recursion (2.3.11) and (2.4.7). Each curve approaches $1/A(p)$. As $p \uparrow 1/2$, both the limiting height and the time needed to approach it increase. The plotted data are generated by `scripts/reproduce_numerics.py`.

2.4.3 The limiting conditional variance

Let $M_n = \mathbb{E}[Z_n^2]$, and write $\sigma^2 = \text{Var}(\xi)$. Conditional on Z_n , the next generation is a sum of Z_n independent offspring counts, so

$$\mathbb{E}[Z_{n+1}^2 \mid Z_n] = \sigma^2 Z_n + m^2 Z_n^2. \quad (2.4.9)$$

Taking expectations and using (2.3.3),

$$M_{n+1} = m^2 M_n + \sigma^2 m^n, \quad M_0 = 1. \quad (2.4.10)$$

For the binary law, $\sigma^2 = 4p(1-p) = m(2-m)$. Solving (2.4.10) for $m \neq 1$ gives

$$\begin{aligned}\mathbb{E}[Z_n^2] &= m^{2n} + \sigma^2 m^{n-1} \frac{1-m^n}{1-m} \\ &= m^n \left(1 + \frac{1}{1-m} \right) - \frac{m^{2n}}{1-m}.\end{aligned}\quad (2.4.11)$$

The second line is specific to the binary variance. It returns $\mathbb{E}[Z_1^2] = 4p$, providing a quick sign check on the final term.

For $m < 1$, division by $S_n \sim A(p)m^n$ yields

$$\mathbb{E}[Z_n^2 | Z_n > 0] = \frac{\mathbb{E}[Z_n^2]}{S_n} \longrightarrow \frac{1}{A(p)} \left(1 + \frac{1}{1-2p} \right). \quad (2.4.12)$$

Combining this with (2.4.8), the conditional variance satisfies

$$\text{Var}(Z_n | Z_n > 0) \longrightarrow \frac{1}{A(p)} \left(1 + \frac{1}{1-2p} \right) - \frac{1}{A(p)^2}. \quad (2.4.13)$$

These limits are the first two moments of the binary Yaglom law. Both become large near criticality, where the conditioning retains increasingly broad populations.

2.4.4 A continuum approximation to the recursion

Although no approximation is needed for the preceding results, a continuum view of the recursion helps to locate the near-critical time scale. Put $\varepsilon = 1 - 2p > 0$. The exact increment is

$$S_{n+1} - S_n = -\varepsilon S_n - p S_n^2. \quad (2.4.14)$$

Replacing the unit difference by a derivative gives the Riccati equation

$$\frac{dS}{dn} = -\varepsilon S - p S^2. \quad (2.4.15)$$

The substitution $v = 1/S$ gives $v' = \varepsilon v + p$, and hence

$$S(n) = \frac{\varepsilon}{C\varepsilon e^{\varepsilon n} - p}, \quad (2.4.16)$$

where C is fixed by matching to the early discrete dynamics. At criticality, (2.4.15) reduces to $S' = -S^2/2$ and recovers the scale $2/n$. For $\varepsilon > 0$, it predicts a crossover on the scale $n \asymp 1/\varepsilon$ followed by exponential decay.

The approximation cannot determine $A(p)$: its integration constant is controlled by precisely those early generations for which replacing a difference by a derivative is least accurate. We therefore use (2.4.16) only as a guide. The exact product and series in section 2.6 recover the missing amplitude and permit a rigorous near-critical analysis.

2.5 Small founding populations and competing jumps

Criticality is an asymptotic classification. A population founded by one or a few individuals must first survive its early generations, and those generations remain highly variable even on the supercritical side of the threshold. This section quantifies that early loss, extends the calculation to k independent founders, and then connects the discrete process to rupture and continuous-time birth–death mechanisms.

2.5.1 Early survival and k founders

For a single founder, the binary recursion

$$u_{n+1} = 1 - p + pu_n^2, \quad S_{n+1} = 2pS_n - pS_n^2 \quad (2.5.1)$$

starts at $u_0 = 0$ and $S_0 = 1$. At criticality the first four extinction probabilities are

$$u_1 = 0.500, \quad u_2 = 0.625, \quad u_3 = 0.6953125, \quad u_4 = 0.7417297 \dots \quad (2.5.2)$$

Equivalently, survival through two generations has probability

$$S_2 = \frac{1}{2} \left(1 - \frac{1}{4} \right) = \frac{3}{8}. \quad (2.5.3)$$

The factor $1/2$ requires the founder to divide, while $1 - 1/4$ requires at least one of its two descendants to divide in the next generation.

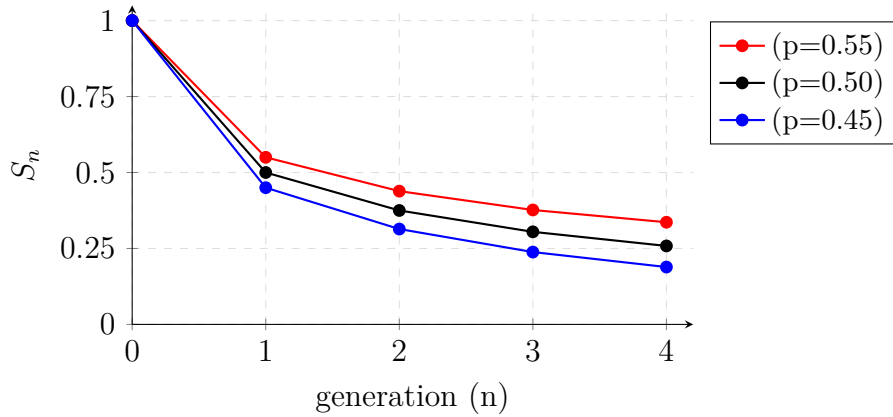


Figure 2.5: Survival over the first four generations from one founder. The early loss is steep in all three regimes; a modest supercritical advantage does not remove the founding bottleneck.

Now let $Z_0 = k$. The k descendant trees are independent, and each is extinct by generation n with probability u_n . The probability that at least one lineage remains is therefore

$$S_n^{(k)} = 1 - u_n^k = 1 - (1 - S_n)^k. \quad (2.5.4)$$

This identity separates two effects: S_n contains the dynamics of one lineage, whereas the exponent k contains the founding size. For small S_n , $S_n^{(k)} \sim kS_n$ when $kS_n \ll 1$, but the exact formula is needed once the survival chance is no longer small.

Conditioning an ensemble on survival also selects realisations with unusually favourable early histories. An apparent early excess of reproduction can therefore arise even when every individual in every realisation obeys the same offspring law. This is a survivorship effect rather than a change of mechanism: selection is being performed on paths after they have been generated. The steep early decline in figure 2.5 is the source of that bias.

2.5.2 A discrete rupture mechanism

The binary process can be augmented by a catastrophe that ends the whole population. The following elementary construction is retained as a bridge to the birth–death–catastrophe models developed later in the thesis; it is not intended as a full theory of those models.

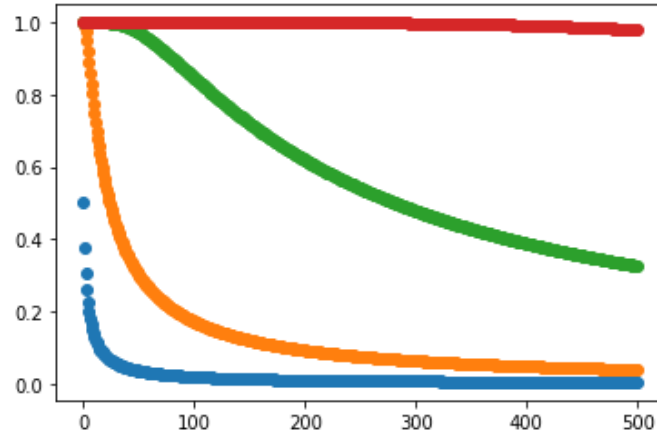
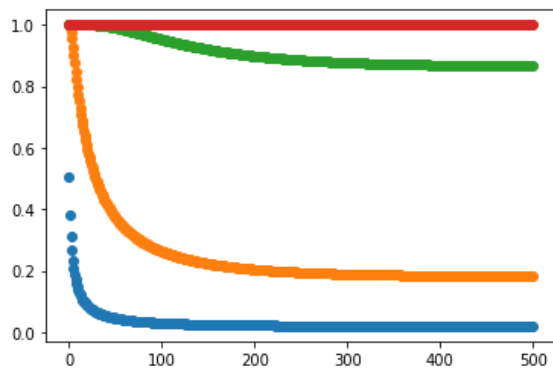
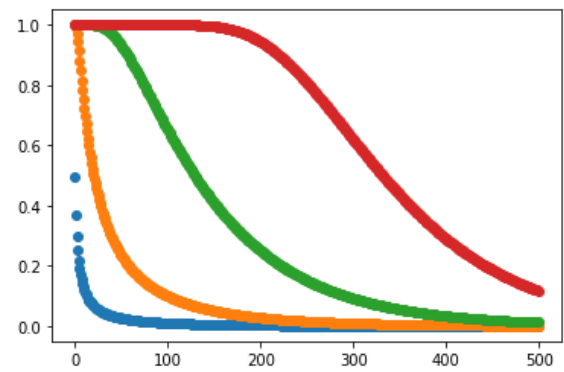


Figure 2.6: Survival probability (2.5.4) through 500 generations at $p = 0.5$, for $k = 1, 10, 100, 1000$. A larger founding population delays extinction, but every curve still tends to zero at criticality.



(a) $p = 0.505$



(b) $p = 0.495$

Figure 2.7: The same founding sizes just above and below criticality. The curves separate slowly at first and then reflect the different limiting survival laws.

Suppose first that the population doubles deterministically at each generation until rupture. Each individual present during a step independently initiates rupture with probability $\kappa \in [0, 1]$, and a single initiation terminates the process. The population exposed during generations $0, \dots, n-1$ has total size

$$1 + 2 + \dots + 2^{n-1} = 2^n - 1. \quad (2.5.5)$$

It follows that the probability of no rupture before generation n is

$$R_n = (1 - \kappa)^{2^n - 1}. \quad (2.5.6)$$

At step n itself, where 2^{n-1} individuals are exposed, the conditional probability of rupture is

$$1 - (1 - \kappa)^{2^{n-1}}. \quad (2.5.7)$$

When death is allowed, the accumulated exposure is random. Let $\tilde{Z}_i$ denote the binary Galton–Watson population with the rupture mechanism switched off, and set

$$H_n = \sum_{i=0}^{n-1} \tilde{Z}_i. \quad (2.5.8)$$

Conditioning on the whole catastrophe-free genealogy gives the exact identity

$$\mathbb{P}(\text{no rupture before } n) = \mathbb{E} \left[(1 - \kappa)^{H_n} \right]. \quad (2.5.9)$$

Since $h \mapsto (1 - \kappa)^h$ is convex, Jensen's inequality supplies the lower bound

$$\mathbb{P}(\text{no rupture before } n) \geq (1 - \kappa)^{\mathbb{E}[H_n]}, \quad (2.5.10)$$

where, with $m = 2p$,

$$\mathbb{E}[H_n] = \sum_{i=0}^{n-1} m^i = \begin{cases} \frac{1 - m^n}{1 - m}, & m \neq 1, \\ n, & m = 1. \end{cases} \quad (2.5.11)$$

Thus replacing the random exposure by its mean is not an identity, but it does have a precise lower-bound interpretation for the event of non-rupture.

Non-rupture should not be confused with population survival. If the requirement is that the population be both unruptured and non-zero at generation n , then the exact probability is

$$\mathbb{E} \left[\mathbf{1}_{\{\tilde{Z}_n > 0\}} (1 - \kappa)^{H_n} \right], \quad (2.5.12)$$

for which (2.5.10) does not directly provide a bound. This distinction removes an ambiguity in the elementary approximation and also indicates why the joint quasi-stationary problem is better posed through the killed-chain generator developed in later chapters.

2.5.3 The linear continuous-time birth–death process

Let $X_t \in \mathbb{Z}_{\geq 0}$ have per-capita birth rate $\lambda > 0$ and death rate $\mu > 0$. From state $i \geq 1$,

$$i \longrightarrow i + 1 \quad \text{at rate } i\lambda, \quad i \longrightarrow i - 1 \quad \text{at rate } i\mu, \quad (2.5.13)$$

with 0 absorbing. The process is the continuous-time branching analogue of (2.3.7). Its extinction law follows from the one-founder backward equation.

Proposition 2.3 (Finite-time extinction). *Starting from N individuals, the probability of extinction by time t is [4]*

$$p_0^{(N)}(t) = \begin{cases} \left[\frac{\mu\{1 - e^{-(\lambda-\mu)t}\}}{\lambda - \mu e^{-(\lambda-\mu)t}} \right]^N, & \lambda \neq \mu, \\ \left(\frac{\lambda t}{1 + \lambda t} \right)^N, & \lambda = \mu. \end{cases} \quad (2.5.14)$$

Proof. Let

$$F(s, t) = \mathbb{E}[s^{X_t} \mid X_0 = 1]. \quad (2.5.15)$$

During an initial interval of length h , one founder dies, gives birth, or experiences no event with probabilities μh , λh and $1 - (\lambda + \mu)h$, up to $o(h)$. After a birth there are two independent descendant populations, so the branching property gives

$$F(s, t + h) = \mu h + \lambda h F(s, t)^2 + \{1 - (\lambda + \mu)h\} F(s, t) + o(h). \quad (2.5.16)$$

Hence

$$\frac{\partial F}{\partial t} = \lambda F^2 - (\lambda + \mu)F + \mu = \lambda(F - 1) \left(F - \frac{\mu}{\lambda} \right), \quad F(s, 0) = s. \quad (2.5.17)$$

The one-founder extinction probability $q(t) = F(0, t)$ therefore solves the same Riccati equation with $q(0) = 0$. Separation gives, for $\lambda \neq \mu$,

$$q(t) = \frac{\mu\{1 - e^{-(\lambda-\mu)t}\}}{\lambda - \mu e^{-(\lambda-\mu)t}}. \quad (2.5.18)$$

When $\lambda = \mu$, the equation is $q' = \lambda(q - 1)^2$, which gives

$$q(t) = \frac{\lambda t}{1 + \lambda t}. \quad (2.5.19)$$

The descendant populations of N founders are independent, and all must be extinct; thus $p_0^{(N)}(t) = q(t)^N$, proving the result. $\square$

The survival probability is $1 - p_0^{(N)}(t)$, while

$$\mathbb{E}[X_t \mid X_0 = N] = N e^{(\lambda-\mu)t}. \quad (2.5.20)$$

For example, differentiating (2.5.17) with respect to s at $s = 1$ gives $M'(t) = (\lambda - \mu)M(t)$, $M(0) = 1$, and hence the displayed mean for one founder; independence supplies the factor N .

For one founder, division of (2.5.20) by the survival probability gives

$$\mathbb{E}[X_t \mid X_t > 0, X_0 = 1] = \frac{\lambda e^{(\lambda-\mu)t} - \mu}{\lambda - \mu}, \quad \lambda \neq \mu. \quad (2.5.21)$$

At $\lambda = \mu$, the corresponding exact conditional mean is

$$\mathbb{E}[X_t \mid X_t > 0, X_0 = 1] = 1 + \lambda t. \quad (2.5.22)$$

In the subcritical case $\mu > \lambda$,

$$\lim_{t \rightarrow \infty} \mathbb{E}[X_t \mid X_t > 0] = \frac{\mu}{\mu - \lambda} = \frac{1}{A_c}. \quad (2.5.23)$$

For fixed $N > 1$, both the mean and the late survival probability acquire the same factor N , so the limit is unchanged. Parameterising the two competing event types by their conditional birth probability

$$p = \frac{\lambda}{\lambda + \mu} \quad (2.5.24)$$

gives

$$A_c(p) = 1 - \frac{\lambda}{\mu} = \frac{1 - 2p}{1 - p}, \quad 0 \leq p < \frac{1}{2}. \quad (2.5.25)$$

Unlike its discrete-time counterpart, A_c is elementary. The comparison between $A_c(p)$ and $A(p)$ will become exact once the series representation of [section 2.6.2](#) is available.

2.5.4 Birth–death with catastrophe as a killed chain

The later birth–death–catastrophe models can be placed in a common framework without introducing their full parameter catalogues here. Let the positive state i have birth rate λ_i , ordinary death rate μ_i , and catastrophe rate κ_i . A catastrophe sends the process directly to 0:

$$\begin{aligned} i &\longrightarrow i + 1 && \text{at rate } \lambda_i, \\ i &\longrightarrow i - 1 && \text{at rate } \mu_i, \\ i &\longrightarrow 0 && \text{at rate } \kappa_i. \end{aligned} \quad (2.5.26)$$

The jump $1 \rightarrow 0$ may therefore occur through ordinary death or catastrophe. On bounded test functions, the generator is

$$\begin{aligned} (\mathcal{L}f)(i) &= \lambda_i \{f(i+1) - f(i)\} + \mu_i \{f(i-1) - f(i)\} \\ &\quad + \kappa_i \{f(0) - f(i)\}, \quad i \geq 1, \end{aligned} \quad (2.5.27)$$

with $(\mathcal{L}f)(0) = 0$. Linear birth and death correspond to $\lambda_i = i\lambda$ and $\mu_i = i\mu$; the catastrophe sequence can remain general.

Restricting the generator to the positive states gives the killed subgenerator K . Its off-diagonal entries are

$$K_{i,i+1} = \lambda_i, \quad K_{i,i-1} = \mu_i \quad (i \geq 2), \quad (2.5.28)$$

and

$$K_{i,i} = -(\lambda_i + \mu_i + \kappa_i). \quad (2.5.29)$$

The missing row mass is exactly the instantaneous absorption rate: it is κ_i for $i \geq 2$ and $\mu_1 + \kappa_1$ for $i = 1$. If $T_0 = \inf\{t \geq 0 : X_t = 0\}$, summing the positive-state forward equations gives

$$\frac{d}{dt} \mathbb{P}(T_0 > t) = -\mu_1 \mathbb{P}(X_t = 1) - \sum_{i \geq 1} \kappa_i \mathbb{P}(X_t = i), \quad (2.5.30)$$

where all probabilities on the right are understood before absorption. Ordinary death removes survival mass only from state 1; catastrophe can remove it from every positive state.

This balance also sharpens the connection with quasi-stationarity. If a distribution $\nu = (\nu_i)_{i \geq 1}$ is quasi-stationary and the relevant sums are finite, then for some $\theta > 0$

$$\nu K = -\theta \nu, \quad \mathbb{P}_\nu(T_0 > t) = e^{-\theta t}. \quad (2.5.31)$$

Indeed, quasi-stationarity writes the unnormalised positive-state law as $a(t)\nu$. The semigroup property forces $a(s+t) = a(s)a(t)$, and right-continuity gives $a(t) = e^{-\theta t}$. Differentiating at zero yields the left eigenvector relation. Summing its coordinates and using (2.5.30) identifies the decay rate:

$$\theta = \mu_1 \nu_1 + \sum_{i \geq 1} \kappa_i \nu_i. \quad (2.5.32)$$

A constant catastrophe clock is a useful limiting case. If $\kappa_i \equiv \kappa$ on the positive states, that clock may be taken independent of the underlying birth–death process. Its killed semigroup is

$$P_t^K = e^{-\kappa t} P_t^{\text{BD}, K}. \quad (2.5.33)$$

The extra clock shifts the survival decay rate by κ but leaves the law conditional on survival unchanged. By contrast, a state-dependent sequence such as $\kappa_i = i\kappa$ weights paths according to their accumulated population exposure and generally changes the quasi-stationary law itself. Later chapters use this killed-generator viewpoint; the present chapter needs only the distinction and the eigenrelation (2.5.31).

2.6 The survival amplitude $A(p)$

For $0 < p < 1/2$, all of the limiting conditional moments obtained in section 2.4 depend on the amplitude

$$A(p) = \lim_{n \rightarrow \infty} \frac{S_n}{(2p)^n}. \quad (2.6.1)$$

Direct iteration gives accurate values away from criticality, but it does not explain the constant and becomes slow as $p \uparrow 1/2$. We now derive an exact product, an exact series, two-sided bounds, a rigorous near-critical asymptotic and a structural connection with linearisation in complex dynamics. These results are specific to the binary process and are therefore proved in full.

2.6.1 An infinite product

Put

$$r = 2p \in (0, 1), \quad w_n = \frac{S_n}{2}. \quad (2.6.2)$$

Then the survival recursion becomes the logistic map

$$w_{n+1} = f_r(w_n) := r w_n (1 - w_n), \quad w_0 = \frac{1}{2}. \quad (2.6.3)$$

Only the attracting regime $0 < r < 1$ is relevant here. Define $x_n = w_n/r^n$. Successive ratios telescope:

$$\frac{x_{n+1}}{x_n} = \frac{w_{n+1}}{rw_n} = 1 - w_n. \quad (2.6.4)$$

Since $A(p) = 2 \lim_n x_n$ and $2x_0 = 1$, this suggests the following exact representation.

Proposition 2.4 (Product representation). *For $0 < p < 1/2$,*

$$A(p) = \prod_{k=0}^{\infty} (1 - w_k) = \frac{1}{2} \prod_{k=1}^{\infty} \left(1 - \frac{S_k}{2}\right). \quad (2.6.5)$$

The product converges to a strictly positive number. In particular, the limit in (2.6.1) exists.

Proof. Iteration of (2.6.4) gives

$$x_n = x_0 \prod_{k=0}^{n-1} (1 - w_k). \quad (2.6.6)$$

Moreover, $0 < w_{n+1} < rw_n$, so $w_n \leq w_0 r^n$. Hence $\sum_n w_n < \infty$. An infinite product with factors $1 - w_n \in (0, 1)$ has a positive limit whenever $\sum_n w_n$ converges. Passing to the limit and using $w_0 = 1/2$ proves both identities in (2.6.5). $\square$

The finite product

$$A_N(p) = \frac{1}{2} \prod_{k=1}^N \left(1 - \frac{S_k}{2}\right) \quad (2.6.7)$$

decreases to $A(p)$, so every truncation is an upper bound. A useful lower bound comes from $\prod_k (1 - a_k) \geq 1 - \sum_k a_k$ for $a_k \in [0, 1]$. Because $w_{k+1} \leq rw_k$,

$$A_N \left(1 - \frac{w_{N+1}}{1 - r}\right) \leq A(p) \leq A_N, \quad (2.6.8)$$

whenever the left factor is positive. The bracket provides a stopping criterion that is stronger than merely observing that successive floating-point iterates have stopped changing.

2.6.2 An exact series and two-sided bounds

The product proves existence, but an additive representation is more effective near criticality. Set

$$v_n = \frac{r^n}{S_n}. \quad (2.6.9)$$

Then $v_n \rightarrow 1/A(p)$, and the factorised recursion $S_{n+1} = rS_n(1 - S_n/2)$ gives

$$\begin{aligned} v_{n+1} - v_n &= \frac{r^{n+1}}{rS_n(1 - S_n/2)} - \frac{r^n}{S_n} \\ &= \frac{r^n}{2 - S_n}. \end{aligned} \quad (2.6.10)$$

Since $v_0 = 1$, summation yields an exact series.

Proposition 2.5 (Series representation). *For $0 < p < 1/2$,*

$$\frac{1}{A(p)} = 1 + \sum_{n=0}^{\infty} \frac{(2p)^n}{2 - S_n}. \quad (2.6.11)$$

The nonlinearity is confined to the denominator, while the numerator is geometric. Since $0 < S_n \leq 1$, we have

$$\frac{1}{2}r^n < \frac{r^n}{2 - S_n} \leq r^n, \quad (2.6.12)$$

with strictness on the appropriate side for at least one term. Write $\varepsilon = 1 - r = 1 - 2p$. Summing the geometric bounds and inverting gives the following result.

Proposition 2.6 (Two-sided bounds). *For $0 < p < 1/2$,*

$$\frac{\varepsilon}{1 + \varepsilon} < A(p) < \frac{2\varepsilon}{1 + 2\varepsilon}. \quad (2.6.13)$$

Equivalently,

$$\frac{1 - 2p}{2(1 - p)} < A(p) < \frac{2(1 - 2p)}{3 - 4p}. \quad (2.6.14)$$

The lower bound is one half of the continuous-time constant:

$$A(p) > \frac{1}{2}A_c(p). \quad (2.6.15)$$

Proof. By (2.6.12),

$$1 + \frac{1}{2\varepsilon} < \frac{1}{A(p)} < 1 + \frac{1}{\varepsilon}. \quad (2.6.16)$$

All quantities are positive, so inversion reverses the inequalities and gives (2.6.13). Substitution of $\varepsilon = 1 - 2p$ gives (2.6.14). Finally, $A_c(p) = 2\varepsilon/(1 + \varepsilon)$ by (2.5.25), proving (2.6.15). $\square$

There is also a bound that comes from the support of the conditional population.

Proposition 2.7 (Parity bound). *For $0 < p < 1/2$,*

$$A(p) \leq \frac{1}{2}, \quad (2.6.17)$$

and $A(p) \rightarrow 1/2$ as $p \downarrow 0$.

Proof. For every $n \geq 1$, the binary process has even population size. Conditional on survival, $Z_n \geq 2$, so (2.4.7) gives $m^n/S_n \geq 2$. Letting $n \rightarrow \infty$ yields $1/A(p) \geq 2$. On the other hand, the lower bound in (2.6.14) tends to $1/2$ as $p \downarrow 0$; together with (2.6.17), this proves the endpoint limit. $\square$

The definition is therefore extended continuously by

$$A(0) := \lim_{p \downarrow 0} A(p) = \frac{1}{2}. \quad (2.6.18)$$

This convention avoids the undefined ratio $S_n/(2p)^n$ at $p = 0$ itself.

Table 2.1: High-precision values of $A(p)$, the bounds in [proposition 2.6](#), the near-critical approximation $2(1 - 2p)$, and the conditional mean $1/A(p)$. The last column is the number of product factors used by the reference calculation. The values are reproduced by `scripts/reproduce_numerics.py`.

p	$A_c/2$	$A(p)$	upper bound	$2(1 - 2p)$	$1/A(p)$	factors
0.05	0.473684	0.486180	0.642857	1.800000	2.0568	14
0.10	0.444444	0.469382	0.615385	1.600000	2.1305	20
0.20	0.375000	0.423895	0.545455	1.200000	2.3591	35
0.30	0.285714	0.353967	0.444444	0.800000	2.8251	62
0.40	0.166667	0.237647	0.285714	0.400000	4.2079	143
0.45	0.090909	0.145069	0.166667	0.200000	6.8933	303
0.48	0.038462	0.068011	0.074074	0.080000	14.7036	786
0.49	0.019608	0.036395	0.038462	0.040000	27.4764	1591
0.499	0.001996	0.003944	0.003984	0.004000	253.519	16095
0.4999	0.000200	0.000399	0.000400	0.000400	2504.65	161156

2.6.3 The near-critical limit

The bounds in [\(2.6.13\)](#) show that $A(p)$ is of order ε , but leave a factor of two unresolved. The exact series determines the leading constant and the order of the first relative correction.

Theorem 2.8 (Near-critical survival amplitude). *As $p \uparrow 1/2$, with $\varepsilon = 1 - 2p \downarrow 0$,*

$$A(p) = 2\varepsilon \left[1 + O\left(\varepsilon \log \frac{1}{\varepsilon}\right) \right]. \quad (2.6.19)$$

In particular,

$$A(p) \sim 2(1 - 2p), \quad \frac{1}{A(p)} \sim \frac{1}{2(1 - 2p)}. \quad (2.6.20)$$

Proof. Write $r = 1 - \varepsilon$. The identity

$$\frac{1}{2 - S_n} = \frac{1}{2} + \frac{S_n}{2(2 - S_n)} \quad (2.6.21)$$

turns [\(2.6.11\)](#) into

$$\frac{1}{A(p)} = 1 + \frac{1}{2\varepsilon} + D_\varepsilon, \quad D_\varepsilon := \sum_{n=0}^{\infty} r^n \frac{S_n}{2(2 - S_n)}. \quad (2.6.22)$$

We require only a logarithmic bound on the remainder. The map $s \mapsto ps(2 - s)$ is increasing in both p and s on the relevant range, so the subcritical survival probability is bounded above by its critical counterpart, $S_n(p) \leq S_n(1/2)$. From [\(2.3.17\)](#),

$$\frac{1}{S_{n+1}(1/2)} - \frac{1}{S_n(1/2)} \geq \frac{1}{2}, \quad (2.6.23)$$

and hence $S_n(1/2) \leq 2/(n + 2)$. Since $2 - S_n \geq 1$,

$$0 \leq D_\varepsilon \leq \frac{1}{2} \sum_{n=0}^{\infty} r^n S_n(p) \leq \sum_{n=0}^{\infty} \frac{r^n}{n + 2}$$

$$= \frac{-\log(1-r) - r}{r^2} = O\left(\log \frac{1}{\varepsilon}\right). \quad (2.6.24)$$

Factoring $1/(2\varepsilon)$ from (2.6.22) gives

$$\frac{1}{A(p)} = \frac{1}{2\varepsilon} [1 + 2\varepsilon\{1 + D_\varepsilon\}]. \quad (2.6.25)$$

The quantity after the leading one is $O(\varepsilon \log(1/\varepsilon))$; inversion proves (2.6.19). $\square$

The proof locates the logarithm that slows convergence to the leading asymptotic. Numerically, the ratios $A/(2\varepsilon)$ are approximately 0.90987, 0.98612, 0.99814, 0.99977, 0.999972 as ε decreases from 2×10^{-2} to 2×10^{-6} . An exploratory fit gives

$$1 - \frac{A}{2\varepsilon} \approx \varepsilon \left(\log \frac{1}{\varepsilon} + 0.78 \right), \quad (2.6.26)$$

but the constant 0.78 is empirical and is not used subsequently. The proved statement is the order estimate in (2.6.19).

2.6.4 Discrete and continuous time

The continuous-time amplitude from (2.5.25) is

$$A_c(p) = \frac{1-2p}{1-p} = \frac{2\varepsilon}{1+\varepsilon}. \quad (2.6.27)$$

The lower bound in (2.6.13) is exactly $A_c/2$. At the strongly subcritical endpoint,

$$\frac{A(p)}{A_c(p)} \longrightarrow \frac{1}{2} \quad (p \downarrow 0), \quad (2.6.28)$$

whereas (2.6.20) and (2.6.27) give

$$\frac{A(p)}{A_c(p)} \longrightarrow 1 \quad (p \uparrow \tfrac{1}{2}). \quad (2.6.29)$$

The factor of two at $p \downarrow 0$ is a support effect. A surviving binary generation contains at least two individuals, while a surviving continuous-time population may contain one. The respective limiting conditional means are therefore two and one. Near criticality this distinction is lower order, and both amplitudes share the leading behaviour 2ε .

Computed values of A/A_c increase from approximately 0.503, 0.528, 0.619, 0.798, 0.928 to 0.988 at $p = 0.01, 0.1, 0.3, 0.45, 0.49, 0.499$, respectively. This numerical monotonicity is descriptive; no monotonicity theorem is required for (2.6.28) and (2.6.29).

2.6.5 Koenigs linearisation

The product in (2.6.5) is also the classical linearising coordinate of the logistic map. For $f_r(z) = rz(1-z)$, $0 < r < 1$, Koenigs' theorem [5] gives a unique function ψ_r , analytic near the attracting fixed point 0, such that

$$\psi_r(f_r(z)) = r\psi_r(z), \quad \psi_r(0) = 0, \quad \psi'_r(0) = 1. \quad (2.6.30)$$

The function extends along the attracting basin by the functional equation. In the ψ_r -coordinate, one nonlinear iteration of f_r is simply multiplication by r .

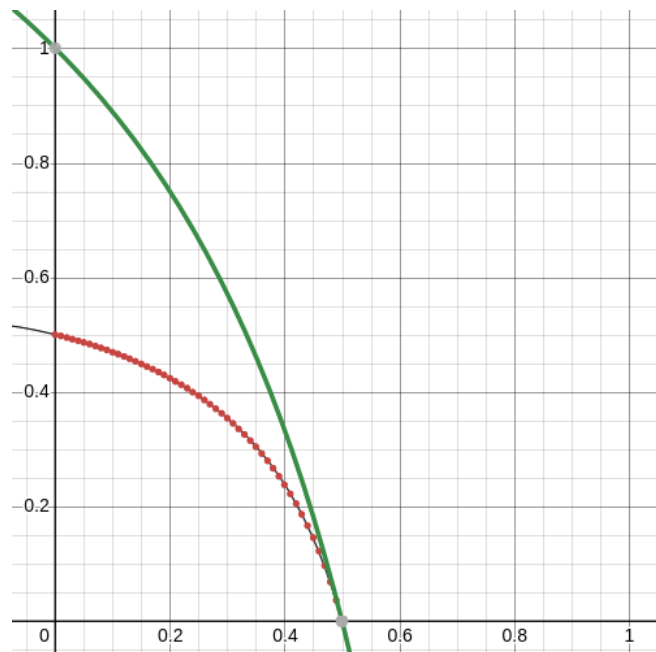


Figure 2.8: The elementary continuous-time amplitude $A_c(p)$ and numerical values of the discrete-time $A(p)$. The additional black curve is an exploratory symbolic-regression approximation discussed in [section 2.A](#); it is visually close but fails the critical asymptotic.

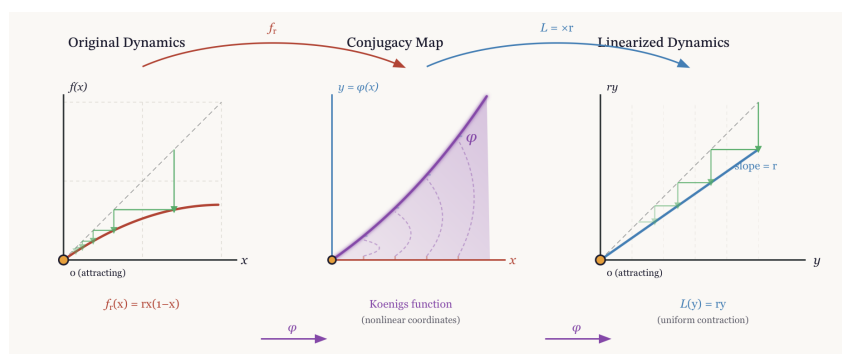


Figure 2.9: Koenigs linearisation. The coordinate change ψ_r conjugates the nonlinear map f_r to multiplication by r , expressing (2.6.30).

Proposition 2.9 (The survival amplitude as a Koenigs value). *For $0 < p < 1/2$ and $r = 2p$,*

$$A(p) = 2\psi_r\left(\frac{1}{2}\right). \quad (2.6.31)$$

Proof. Let $w_{n+1} = f_r(w_n)$ with $w_0 = 1/2$. Iterating (2.6.30) gives

$$\psi_r(w_0) = \frac{\psi_r(w_n)}{r^n}. \quad (2.6.32)$$

Since $w_n \rightarrow 0$ and $\psi_r(z)/z \rightarrow 1$ as $z \rightarrow 0$,

$$\psi_r(w_0) = \lim_{n \rightarrow \infty} \frac{w_n}{r^n}. \quad (2.6.33)$$

But $w_n = S_n/2$, so multiplication by two gives (2.6.31). $\square$

This identification explains why elementary curve fitting is an unreliable route to $A(p)$. Becker and Bergweiler’s classification [6] shows that Schröder conjugacies of rational maps are hypertranscendental outside a short list of exceptional dynamical families. The quadratic maps corresponding to $r \in (0, 1)$ are not exceptional; the conjugacy calculation is recorded in section 2.A.2. Thus, for each fixed $r \in (0, 1)$, the function $z \mapsto \psi_r(z)$ satisfies no algebraic differential equation over the rational functions and, in particular, is not elementary.

The conclusion must stop there. Hypertranscendence in the argument z does not prove that the parameter map

$$p \longmapsto 2\psi_{2p}(1/2) = A(p) \quad (2.6.34)$$

is non-elementary, because this map varies r while fixing the argument. No elementary expression for $A(p)$ is known, and the Koenigs representation gives a structural reason not to expect one, but it is not a transcendence theorem for the values in (2.6.34). The exploratory searches and the conjugacy to the quadratic family are recorded, with this limitation, in section 2.A.

2.6.6 Computation in practice

Away from criticality, the product (2.6.7) is direct, monotone and accompanied by the certified bracket (2.6.8). Near criticality the number of factors grows on the scale $1/(1-2p)$, with an additional logarithmic cost if many digits are required. The final column of table 2.1 shows this slowdown.

For calculations that do not require a certified product value arbitrarily close to criticality, the exact product and the asymptotic can be combined:

$$A(p) \approx \begin{cases} \frac{1}{2} \prod_{k=1}^N \left(1 - \frac{S_k}{2}\right), & p < c, \\ 2(1-2p), & c \leq p < \frac{1}{2}. \end{cases} \quad (2.6.35)$$

Here N is chosen from (2.6.8), and c is selected to match the required accuracy. The relative error of the second branch is $O(\{1-2p\} \log\{1/(1-2p)\})$ by theorem 2.8. For example, Choosing $c = 0.4999$ gives a sub-percent approximation across the whole interval;

$c = 0.499$ leaves an error of about 1.4% at the join. A smooth blend can be used if a differentiable surrogate is needed.

All values in this chapter are generated from the product rather than from a fitted formula. The accompanying script uses decimal arithmetic, reports the rigorous tail bracket and switches to the asymptotic only when explicitly requested.

2.7 Absorption models and the method of characteristics

We now consider a population split between the exterior and interior of a single compartment. Transfer moves an exterior particle to the interior; interior particles may then die or reproduce. The models are deliberately elementary, but their two-dimensional state and linear per-capita rates already produce the generating-function PDEs needed later for systems with rupture.

Let X_t and Y_t denote the interior and exterior counts, respectively, and write

$$p_{i,j}(t) = \mathbb{P}(X_t = i, Y_t = j), \quad G(x, y, t) = \mathbb{E}[x^{X_t} y^{Y_t}] = \sum_{i,j \geq 0} p_{i,j}(t) x^i y^j. \quad (2.7.1)$$

Every model starts from $X_0 = 0, Y_0 = N$, and hence

$$G(x, y, 0) = y^N. \quad (2.7.2)$$

Three cases are treated in increasing order of difficulty. Transfer alone and transfer followed by death have finite reachable state spaces and admit direct single-particle solutions. Adding birth opens an infinite state space. The branching property still raises the one-founder generating function to the power N , but it no longer supplies that one-founder function; the PDE must then be solved.

2.7.1 Transfer only

Each exterior particle transfers at rate $\alpha > 0$, so the only jump is

$$(i, j) \longrightarrow (i + 1, j - 1) \quad \text{at rate } \alpha j. \quad (2.7.3)$$

Equivalently, over a short interval h ,

$$\begin{aligned} \mathbb{P}(\Delta X = 1, \Delta Y = -1 \mid X_t = i, Y_t = j) &= \alpha j h + o(h), \\ \mathbb{P}(\Delta X = 0, \Delta Y = 0 \mid X_t = i, Y_t = j) &= 1 - \alpha j h + o(h). \end{aligned} \quad (2.7.4)$$

The componentwise forward equation is

$$\frac{dp_{i,j}}{dt} = \alpha(j+1)p_{i-1,j+1} - \alpha j p_{i,j}. \quad (2.7.5)$$

The generating-function PDE follows directly from the generator. Applied to the monomial $f(i, j) = x^i y^j$,

$$\mathcal{L}f(i, j) = \alpha j \{x^{i+1} y^{j-1} - x^i y^j\}$$

$$= \alpha(x - y) \frac{\partial}{\partial y} (x^i y^j). \quad (2.7.6)$$

Taking expectations gives

$$\frac{\partial G}{\partial t} = \alpha(x - y) \frac{\partial G}{\partial y}, \quad G(x, y, 0) = y^N. \quad (2.7.7)$$

This generator calculation is equivalent to multiplying (2.7.5) by $x^i y^j$, summing and shifting indices, but it keeps the jump that produces each derivative visible.

The direct solution uses independence. For one initial exterior particle the only states are $(0, 1)$ and $(1, 0)$, with

$$p_{0,1}(t) = e^{-\alpha t}, \quad p_{1,0}(t) = 1 - e^{-\alpha t}. \quad (2.7.8)$$

Thus

$$G_1(x, y, t) = (1 - e^{-\alpha t})x + e^{-\alpha t}y. \quad (2.7.9)$$

The N initial particles move independently, so

$$G_N(x, y, t) = \left\{ (1 - e^{-\alpha t})x + e^{-\alpha t}y \right\}^N. \quad (2.7.10)$$

In particular,

$$X_t \sim \text{Binomial}(N, 1 - e^{-\alpha t}), \quad Y_t = N - X_t. \quad (2.7.11)$$

Direct differentiation confirms that (2.7.10) satisfies (2.7.7).

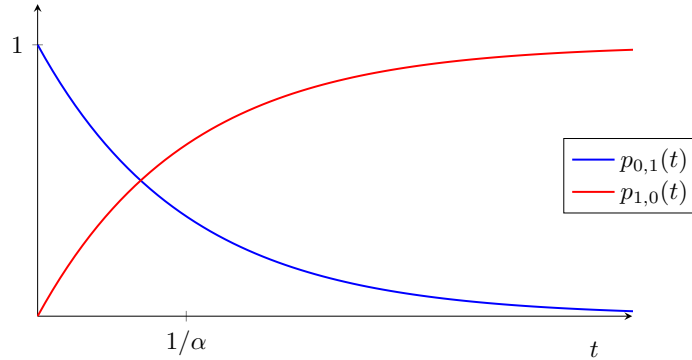


Figure 2.10: State probabilities for one particle in the transfer-only model. The probability of remaining outside decays at rate α ; the complementary probability records transfer to the interior.

2.7.2 Transfer followed by death

Now let each interior particle die at rate $\mu > 0$. The jumps are

$$\begin{aligned} (i, j) &\longrightarrow (i + 1, j - 1) && \text{at rate } \alpha j, \\ (i, j) &\longrightarrow (i - 1, j) && \text{at rate } \mu i. \end{aligned} \quad (2.7.12)$$

The forward equations become

$$\frac{dp_{i,j}}{dt} = \alpha(j + 1)p_{i-1,j+1} + \mu(i + 1)p_{i+1,j} - (\alpha j + \mu i)p_{i,j}. \quad (2.7.13)$$

Using the two jump increments in the generator gives

$$\frac{\partial G}{\partial t} = \mu(1-x)\frac{\partial G}{\partial x} + \alpha(x-y)\frac{\partial G}{\partial y}, \quad G(x, y, 0) = y^N. \quad (2.7.14)$$

For one initial particle there are three states: exterior $(0, 1)$, interior $(1, 0)$, and dead $(0, 0)$. The first two probabilities solve

$$p'_{0,1} = -\alpha p_{0,1}, \quad p'_{1,0} = \alpha p_{0,1} - \mu p_{1,0}. \quad (2.7.15)$$

When $\alpha \neq \mu$,

$$\begin{aligned} p_{0,1}(t) &= e^{-\alpha t}, \\ p_{1,0}(t) &= \frac{\alpha}{\mu - \alpha}(e^{-\alpha t} - e^{-\mu t}), \\ p_{0,0}(t) &= 1 - p_{0,1}(t) - p_{1,0}(t). \end{aligned} \quad (2.7.16)$$

At the removable degeneracy $\alpha = \mu$, the middle expression is replaced by

$$p_{1,0}(t) = \alpha t e^{-\alpha t}. \quad (2.7.17)$$

Independence of the initial particles again gives a power:

$$G_N(x, y, t) = \{p_{0,0}(t) + p_{1,0}(t)x + p_{0,1}(t)y\}^N. \quad (2.7.18)$$

For $\alpha \neq \mu$, this is

$$G_N(x, y, t) = \left\{ 1 - \frac{\mu}{\mu - \alpha}e^{-\alpha t} + \frac{\alpha}{\mu - \alpha}e^{-\mu t} + \frac{\alpha}{\mu - \alpha}(e^{-\alpha t} - e^{-\mu t})x + e^{-\alpha t}y \right\}^N. \quad (2.7.19)$$

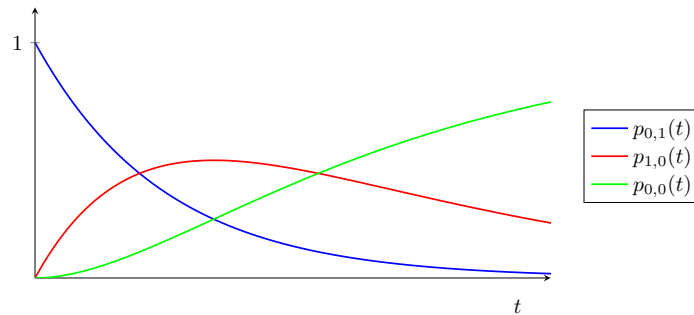


Figure 2.11: State probabilities for one particle under transfer followed by death. The interior probability rises while transfer dominates and then decays as the absorbing dead state accumulates probability.

2.7.3 Characteristics tested on the transfer–death PDE

The direct calculation has already supplied (2.7.19). This makes (2.7.14) a useful test problem for the method of characteristics: the method can be developed without leaving its answer unchecked.

Introduce parameters (s, q, τ) and write

$$(x, y, t) = (x(s, q, \tau), y(s, q, \tau), t(s, q, \tau)), \quad (2.7.20)$$

with initial surface

$$x(s, q, 0) = s, \quad y(s, q, 0) = q, \quad t(s, q, 0) = 0. \quad (2.7.21)$$

Along a curve with s and q fixed, the chain rule gives

$$\frac{dG}{d\tau} = G_t \frac{dt}{d\tau} + G_x \frac{dx}{d\tau} + G_y \frac{dy}{d\tau}. \quad (2.7.22)$$

Writing (2.7.14) as $-G_t + \mu(1-x)G_x + \alpha(x-y)G_y = 0$ and matching coefficients gives

$$\frac{dG}{d\tau} = 0, \quad \frac{dt}{d\tau} = -1, \quad \frac{dx}{d\tau} = \mu(1-x), \quad \frac{dy}{d\tau} = \alpha(x-y). \quad (2.7.23)$$

The first two equations and (2.7.21) give

$$G(s, q, \tau) = q^N, \quad t = -\tau. \quad (2.7.24)$$

Solving the x -equation gives

$$1-x = (1-s)e^{-\mu\tau}. \quad (2.7.25)$$

Substitution into the final equation and use of an integrating factor yield, for $\alpha \neq \mu$,

$$y = 1 - \frac{\alpha}{\alpha - \mu}(1-s)e^{-\mu\tau} + \left\{ q + \frac{\alpha}{\alpha - \mu}(1-s) - 1 \right\} e^{-\alpha\tau}. \quad (2.7.26)$$

The equal-rate case follows by taking the limit $\mu \rightarrow \alpha$.

It remains to return from (s, q, τ) to (x, y, t) . Rearranging (2.7.26) for q , then substituting $1-s = (1-x)e^{\mu\tau}$ from (2.7.25) and $\tau = -t$, gives

$$q = 1 - \frac{\mu}{\mu - \alpha}e^{-\alpha t} + \frac{\alpha}{\mu - \alpha}e^{-\mu t} + \frac{\alpha}{\mu - \alpha}(e^{-\alpha t} - e^{-\mu t})x + e^{-\alpha t}y. \quad (2.7.27)$$

Finally $G = q^N$ by (2.7.24), reproducing (2.7.19). The characteristic orientation, signs and initial surface have therefore all been checked against an independent solution.

2.7.4 Transfer, birth and death

Add linear birth of interior particles at per-capita rate $\beta > 0$. The jump rates are now

$$\begin{aligned} (i, j) &\longrightarrow (i+1, j-1) && \text{at rate } \alpha j, \\ (i, j) &\longrightarrow (i+1, j) && \text{at rate } \beta i, \\ (i, j) &\longrightarrow (i-1, j) && \text{at rate } \mu i. \end{aligned} \quad (2.7.28)$$

The corresponding forward equations are

$$\begin{aligned} \frac{dp_{i,j}}{dt} = & \alpha(j+1)p_{i-1,j+1} + \mu(i+1)p_{i+1,j} + \beta(i-1)p_{i-1,j} \\ & - (\alpha j + \mu i + \beta i)p_{i,j}. \end{aligned} \quad (2.7.29)$$

The generator applied to $x^i y^j$ gives the first-order PDE

$$\frac{\partial G}{\partial t} = \{\mu - (\mu + \beta)x + \beta x^2\} \frac{\partial G}{\partial x} + \alpha(x - y) \frac{\partial G}{\partial y}, \quad G(x, y, 0) = y^N. \quad (2.7.30)$$

Birth makes infinitely many interior states reachable from one transferred particle. Independence still implies $G_N = G_1^N$, but G_1 is now an infinite series and must itself be determined. We solve (2.7.30) directly.

2.7.4.1 The characteristic system

Use the same initial surface as in (2.7.21). The characteristic equations are

$$\begin{aligned} \frac{dG}{d\tau} &= 0, & \frac{dt}{d\tau} &= -1, \\ \frac{dx}{d\tau} &= \mu - (\mu + \beta)x + \beta x^2, & \frac{dy}{d\tau} &= \alpha(x - y). \end{aligned} \quad (2.7.31)$$

Thus $G = q^N$ and $t = -\tau$. The x -equation factorises without a quadratic surd:

$$\mu - (\mu + \beta)x + \beta x^2 = \beta(x - 1)(x - \rho), \quad \rho := \frac{\mu}{\beta}. \quad (2.7.32)$$

Put

$$b := \beta - \mu = \beta(1 - \rho). \quad (2.7.33)$$

When $b \neq 0$, separation of variables and the condition $x(s, q, 0) = s$ give

$$\frac{x - 1}{x - \rho} = \frac{s - 1}{s - \rho} e^{b\tau}. \quad (2.7.34)$$

Writing

$$Z = \frac{s - 1}{s - \rho}, \quad \zeta(\tau) = Z e^{b\tau}, \quad (2.7.35)$$

we may solve (2.7.34) as

$$x(\tau) = \frac{1 - \rho \zeta(\tau)}{1 - \zeta(\tau)} = 1 + (1 - \rho) \frac{\zeta(\tau)}{1 - \zeta(\tau)}. \quad (2.7.36)$$

The last characteristic equation is linear once $x(\tau)$ is known:

$$\frac{d}{d\tau} \{e^{\alpha\tau} y(\tau)\} = \alpha e^{\alpha\tau} x(\tau). \quad (2.7.37)$$

For $|\zeta| < 1$, expand the last quotient in (2.7.36). Termwise integration gives

$$\Psi(\zeta) := 1 + \frac{\alpha}{\beta} \sum_{n=1}^{\infty} \frac{\zeta^n}{n + \delta}, \quad \delta := \frac{\alpha}{b}, \quad (2.7.38)$$

for non-resonant parameter values. Indeed,

$$\frac{d}{d\tau}\{e^{\alpha\tau}\Psi(\zeta(\tau))\} = \alpha e^{\alpha\tau}x(\tau), \quad (2.7.39)$$

because $\alpha + nb = b(n + \delta)$ and $b/\beta = 1 - \rho$. Applying $y(s, q, 0) = q$ to (2.7.37) therefore yields

$$q = e^{\alpha\tau}\{y - \Psi(\zeta(\tau))\} + \Psi(Z). \quad (2.7.40)$$

To return to the original variables, set

$$W = \frac{x - 1}{x - \rho}. \quad (2.7.41)$$

Then $\zeta(\tau) = W$, $Z = We^{bt}$ and $\tau = -t$. Since $G = q^N$, we obtain the promised generating function.

Proposition 2.10 (Transfer–birth–death generating function). *Suppose $\beta \neq \mu$, and define b, ρ, δ, W and Ψ by (2.7.33), (2.7.38) and (2.7.41). Wherever the displayed series converges, and elsewhere by continuation along the characteristic,*

$$\boxed{G(x, y, t) = \left[e^{-\alpha t}\{y - \Psi(W)\} + \Psi(We^{bt}) \right]^N.} \quad (2.7.42)$$

At resonant parameter values the right-hand side is understood by its finite limit, described below.

Proof. Equations (2.7.31) and (2.7.40) show that the quantity inside square brackets is the initial coordinate q of the characteristic through (x, y, t) . The initial condition is $G = q^N$, which proves (2.7.42). The derivation also proves that it satisfies (2.7.30); no uniqueness beyond the usual local uniqueness of the characteristic initial-value problem is being assumed. $\square$

2.7.4.2 A parameter-regular representation

The special-function form is compact, but the underlying probability calculation provides a representation that remains transparent at $\beta = \mu$, at resonance and across hypergeometric branch cuts. Let $F(x, t)$ be the probability generating function at time t of an interior linear birth–death process started from one particle. Its backward equation is

$$\frac{\partial F}{\partial t} = \beta F^2 - (\beta + \mu)F + \mu, \quad F(x, 0) = x. \quad (2.7.43)$$

For $\beta \neq \mu$,

$$\frac{F(x, t) - 1}{F(x, t) - \rho} = \frac{x - 1}{x - \rho} e^{bt}, \quad (2.7.44)$$

while at $\beta = \mu$,

$$F(x, t) = 1 - \frac{1 - x}{1 + \beta t(1 - x)}. \quad (2.7.45)$$

For one initial exterior particle, either transfer has not occurred by time t , or it occurs at time $u \in (0, t)$ and leaves an interior birth–death process of duration $t - u$. Conditioning on that transfer time gives

$$g(x, y, t) = e^{-\alpha t}y + \alpha e^{-\alpha t} \int_0^t e^{\alpha v} F(x, v) dv. \quad (2.7.46)$$

Consequently,

$$G(x, y, t) = g(x, y, t)^N. \quad (2.7.47)$$

This formula is real and finite for $x, y \in [0, 1]$ and all positive rates. For $\beta \neq \mu$, substituting (2.7.44) into the integral and using (2.7.39) recovers (2.7.42). Thus (2.7.46) also specifies unambiguously which continuation of the special-function expression is intended.

Several checks now require almost no algebra. At $t = 0$, (2.7.46) gives $g = y$, and hence $G = y^N$. At $x = y = 1$, $F(1, v) = 1$, so $g = 1$ and total probability is conserved. Setting $x = 1$ gives

$$G(1, y, t) = \{e^{-\alpha t}y + 1 - e^{-\alpha t}\}^N, \quad (2.7.48)$$

which is the generating function of $Y_t \sim \text{Binomial}(N, e^{-\alpha t})$. The exterior marginal is therefore unaffected by the interior birth and death rates, as the transition diagram requires.

2.7.4.3 Hypergeometric form, domains and resonance

The series in (2.7.38) is a Gauss hypergeometric function. With

$${}_2F_1(a, b; c; z) = \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{(c)_n} \frac{z^n}{n!}, \quad (2.7.49)$$

the Pochhammer cancellation

$$\frac{(\delta)_n}{(\delta+1)_n} = \frac{\delta}{\delta+n} \quad (2.7.50)$$

gives

$${}_2F_1(1, \delta; \delta+1; z) = \sum_{n=0}^{\infty} \frac{\delta}{\delta+n} z^n. \quad (2.7.51)$$

It follows that

$$\Psi(z) = 1 + (1 - \rho)\{{}_2F_1(1, \delta; \delta+1; z) - 1\}. \quad (2.7.52)$$

A second derivation, based on a general exponential-rational integral, is given in [section 2.B](#).

The elementary series (2.7.38) converges for $|z| < 1$. If $\mu > \beta$, then $b < 0$, $\rho > 1$, and, for $x \in [0, 1]$,

$$0 \leq W \leq \frac{\beta}{\mu} < 1, \quad 0 \leq W e^{bt} \leq W. \quad (2.7.53)$$

Except at the resonances described next, the series therefore covers the whole probability generating function interval. If $\beta > \mu$, the arguments may leave the unit disc or cross the standard hypergeometric cut. One may then use analytic continuation, tracking the branch along the characteristic, but the real integral (2.7.46) is usually safer. The apparent singularity of W at $x = \rho$ is also only a coordinate singularity; $F(\rho, t) = \rho$ and the continuous limit of (2.7.46) is regular.

When

$$\alpha = n(\mu - \beta), \quad n = 1, 2, \dots, \quad (2.7.54)$$

we have $\delta = -n$, and one denominator in (2.7.38) vanishes. The separate Ψ -terms in (2.7.42) then have poles, but their combination does not. In the resonant term,

$$\frac{(W e^{bt})^n - e^{-\alpha t} W^n}{n + \delta} \quad (2.7.55)$$

has a finite parameter limit and produces the familiar factor $te^{-\alpha t}$. Equivalently, (2.7.46) contains no singular denominator and supplies the limit directly. Resonance is therefore a defect of the separated antiderivative, not of the generating function.

2.7.5 Recovering state probabilities

A generating function is useful only insofar as its coefficients can be recovered. In every model above,

$$p_{i,j}(t) = [x^i y^j]G(x, y, t) = \frac{1}{i! j!} \left. \frac{\partial^{i+j} G}{\partial x^i \partial y^j} \right|_{x=y=0}. \quad (2.7.56)$$

For transfer–birth–death, write

$$h(x, t) := \alpha e^{-\alpha t} \int_0^t e^{\alpha v} F(x, v) \, dv, \quad g(x, y, t) = e^{-\alpha t} y + h(x, t). \quad (2.7.57)$$

The finite exterior expansion then separates the two coefficient extractions:

$$p_{i,j}(t) = \binom{N}{j} e^{-\alpha t j} [x^i] h(x, t)^{N-j}, \quad 0 \leq j \leq N. \quad (2.7.58)$$

The remaining univariate coefficient can be obtained by differentiation, by convolution of truncated series, or by Cauchy’s integral formula. The full coefficient calculations for all three models are collected in [section 2.C](#); the accompanying script implements the convolution route and checks normalisation.

Chapter-local appendices

2.A Closed-form searches and quadratic conjugacy

This appendix records the negative and structural calculations that are useful for interpreting $A(p)$, but are not needed for its probabilistic role in the main text. The numerical formulae below are approximations only. They are retained because they show how little a visually accurate fit says about the singular quantity $1/A(p)$.

2.A.1 Exploratory formulae

Polynomial and rational searches can be constrained to the known endpoints $A(0) = 1/2$ and $A(1/2) = 0$. One of the more economical rational candidates was

$$\hat{A}_1(p) = \frac{\frac{1}{2}(1-2p)(1+2p)}{1 + \frac{1}{2}p - \frac{5}{2}p^2 - \frac{6}{5}p^3 + \frac{6}{5}p^4}. \quad (2.A.1)$$

It passes through both endpoints. At the critical endpoint, however,

$$\hat{A}'_1(1/2) = -\frac{2}{0.55} \approx -3.636, \quad (2.A.2)$$

whereas [theorem 2.8](#) requires the left derivative -4 . Matching the values but missing this slope is enough to make the relative error in A , and hence the error in $1/A$, unacceptable near criticality.

A less restricted symbolic tree search produced many longer formulae of similar visual quality. Two representative outputs were

$$\begin{aligned} \hat{A}_2(p) = & 0.31696448624690005 \log \left(\left| \cos \left(\left(\frac{1}{3.19801133824532} \right)^{1/4} \right. \right. \right. \\ & \left. \left. \left. + p \left[\sin \left(\cos(|p|(0.621865696665606 - p) \cos(e^p)) \right) + |p| \right] \right) \right| \right) + 0.5982282661889831 \end{aligned} \quad (2.A.3)$$

and

$$\hat{A}_3(p) = \frac{-0.616647881739505 e^{-0.552605335919693 e^p}}{\sin(\cos p) - |p|} + 0.921964323679771. \quad (2.A.4)$$

These expressions have no inferential status: they are records of a curve-fitting exercise, not proposed identities.

One can force the correct critical branch by splicing:

$$\hat{A}_4(p) = \begin{cases} -0.04011483983619545 e^{e^{2p}} + 0.6079994336528085, & p < 0.499962, \\ 2 - 4p, & p \geq 0.499962. \end{cases} \quad (2.A.5)$$

This is a practical surrogate, not a closed form for A . If a splice is needed, the certified product in [\(2.6.7\)](#) is preferable on the first branch and the proved asymptotic [\(2.6.19\)](#) on the second.

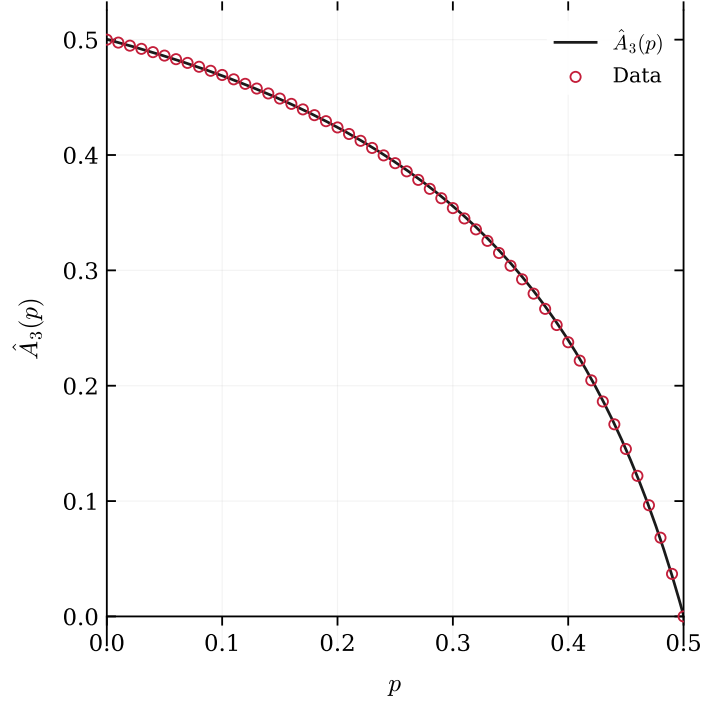


Figure 2.12: The fitted curve $\hat{A}_3$ and product evaluations of $A(p)$. The discrepancy is barely visible on this scale, although $\hat{A}_3(1/2) \approx 9.1 \times 10^{-4}$ rather than zero and its endpoint slope is approximately -3.63 rather than -4 .

2.A.2 Conjugacy with the quadratic family

The logistic family $f_r(x) = rx(1 - x)$ is affinely conjugate to the quadratic family

$$P_c(z) = z^2 + c. \quad (2.A.6)$$

Set

$$z = r \left(\frac{1}{2} - x \right). \quad (2.A.7)$$

If $x' = f_r(x)$ and $z' = r(1/2 - x')$, direct substitution gives

$$z' = z^2 + c, \quad c = c(r) := \frac{2r - r^2}{4}. \quad (2.A.8)$$

The subcritical branching window $0 < r < 1$ therefore maps onto $0 < c < 1/4$.

The interval $0 < c < 1/4$ is the real radius of the main cardioid of the Mandelbrot set. Indeed, a fixed point of $z^2 + c$ with multiplier r has coordinate $z_* = r/2$, and its fixed-point equation gives

$$c = z_* - z_*^2 = \frac{r}{2} - \frac{r^2}{4}. \quad (2.A.9)$$

Thus the endpoint $r \uparrow 1$ approaches the parabolic cusp $c = 1/4$. This observation locates the parameter range; it is not used as evidence for a closed-form claim.

2.A.3 The two elementary comparison cases

The exceptional quadratic maps conjugate to a monomial or Chebyshev polynomial correspond to elementary linearisers. They furnish useful algebraic checks, but neither belongs to $0 < r < 1$.

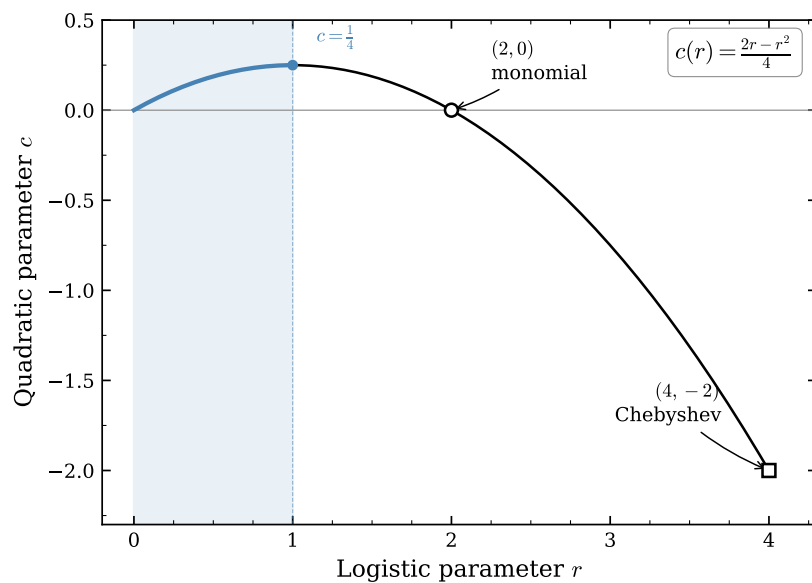


Figure 2.13: The parameter correspondence $c(r) = (2r - r^2)/4$. The subcritical window $0 < r < 1$ maps to $0 < c < 1/4$; the elementary monomial and Chebyshev parameters described below lie outside that window.

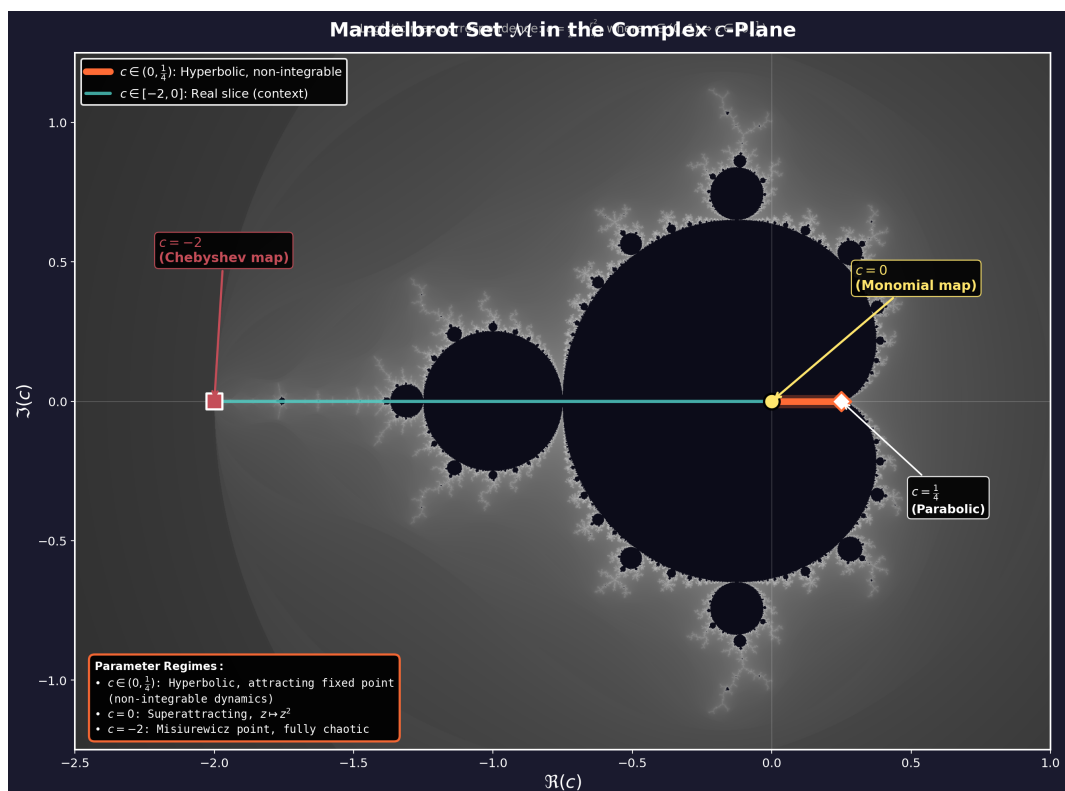


Figure 2.14: The image $0 < c < 1/4$ of the subcritical branching interval within the quadratic parameter plane. The exceptional parameters $c = 0$ and $c = -2$ are shown for comparison.

For $r = 2$, put $u = 1 - 2x$. Then

$$1 - 2f_2(x) = (1 - 2x)^2 = u^2, \quad (2.A.10)$$

and the normalised local lineariser is

$$\psi_2(x) = -\frac{1}{2} \log(1 - 2x). \quad (2.A.11)$$

It satisfies $\psi_2(f_2(x)) = 2\psi_2(x)$, $\psi_2(0) = 0$ and $\psi_2'(0) = 1$.

For $r = 4$, write $x = \sin^2 \theta$. The double-angle identity gives

$$f_4(x) = 4x(1 - x) = \sin^2(2\theta), \quad (2.A.12)$$

so that

$$\psi_4(x) = \{\arcsin(\sqrt{x})\}^2 \quad (2.A.13)$$

satisfies $\psi_4(f_4(x)) = 4\psi_4(x)$ and $\psi_4(x) \sim x$ at the origin.

There is a categorical distinction here. At $r = 2$ and $r = 4$, the origin has multiplier greater than one and is repelling; the formulae (2.A.11) and (2.A.13) are local repelling linearisers. In the branching window, the origin is attracting and the function in (2.6.30) is the attracting Koenigs coordinate. The elementary formulae are therefore comparisons, not analytic continuations of $A(p)$.

Finally, the theorem cited in section 2.6.5 concerns the function $z \mapsto \psi_r(z)$ at fixed r . It does not prove that the parameter evaluation $p \mapsto 2\psi_{2p}(1/2)$ is non-elementary. This logical separation is the material correction to the stronger claim suggested by the original closed-form search.

2.B An exponential–rational hypergeometric integral

The geometric-series derivation of Ψ in section 2.7.4 is the shortest route to the generating function. Direct integration of the rational characteristic leads to the following more general identity.

Proposition 2.11. *Let h, k, p, q be real constants with $hk \neq 0$, $p \neq 0$ and $h/k > 0$. On any interval on which the denominator is non-zero,*

$$\int \frac{e^{h\tau}}{p + qe^{k\tau}} d\tau = \frac{e^{h\tau}}{ph} {}_2F_1\left(1, \frac{h}{k}; \frac{h}{k} + 1; -\frac{q}{p}e^{k\tau}\right) + C, \quad (2.B.1)$$

where the hypergeometric function is continued from its principal power series along the interval. In particular, the Euler-integral proof below applies directly while $-qe^{k\tau}/p \notin [1, \infty)$.

Proof. Use $v = e^{k\tau}$, so that $d\tau = dv/(kv)$. Then

$$\int \frac{e^{h\tau}}{p + qe^{k\tau}} d\tau = \frac{1}{pk} \int \frac{v^{h/k-1}}{1 + (q/p)v} dv. \quad (2.B.2)$$

Write $a = h/k > 0$. An antiderivative may be fixed, up to a constant, by integrating from 0 to v :

$$\frac{1}{pk} \int_0^v \frac{w^{a-1}}{1 + (q/p)w} dw. \quad (2.B.3)$$

The lower endpoint is integrable precisely because $a > 0$. Rescale $w = vu$ to obtain

$$\frac{v^a}{pk} \int_0^1 u^{a-1} \{1 - zu\}^{-1} du, \quad z = -\frac{q}{p}v. \quad (2.B.4)$$

Euler's integral representation states that, for $\Re(c) > \Re(b) > 0$,

$${}_2F_1(a_0, b; c; z) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 u^{b-1} (1-u)^{c-b-1} (1-zu)^{-a_0} du. \quad (2.B.5)$$

Apply it with $a_0 = 1$, $b = a$ and $c = a + 1$. Since $\Gamma(a+1) = a\Gamma(a)$ and $\Gamma(1) = 1$,

$$\int_0^1 \frac{u^{a-1}}{1-zu} du = \frac{1}{a} {}_2F_1(1, a; a+1; z). \quad (2.B.6)$$

Substituting this into (2.B.4), and using $ka = h$, gives

$$\frac{v^a}{ph} {}_2F_1\left(1, \frac{h}{k}; \frac{h}{k} + 1; -\frac{q}{p}v\right). \quad (2.B.7)$$

Finally $v^a = e^{h\tau}$, which proves (2.B.1) on the Euler domain. Both sides are analytic antiderivatives away from the singular set, so continuation along any admissible interval preserves the identity. $\square$

Application to the characteristic

For $\beta \neq \mu$, write the interior characteristic as

$$x(\tau) = 1 + (1 - \rho) \frac{Ze^{b\tau}}{1 - Ze^{b\tau}}, \quad \rho = \frac{\mu}{\beta}, \quad b = \beta - \mu. \quad (2.B.8)$$

The first term integrates elementarily in $\alpha \int e^{\alpha\tau} x(\tau) d\tau$. Away from $\alpha + b = 0$, the second is an instance of [proposition 2.11](#) after taking $h = \alpha + b$, $k = b$, $p = 1$ and $q = -Z$, together with a factor $\alpha(1 - \rho)Z$. The Euler-domain calculation applies when its stated hypotheses hold; elsewhere the identity is continued along the regular characteristic. Algebraic simplification by (2.7.50) returns

$$\alpha \int e^{\alpha\tau} x(\tau) d\tau = e^{\alpha\tau} \Psi(Ze^{b\tau}) + C. \quad (2.B.9)$$

This agrees with (2.7.39). The series proof in the main text has the advantage that the cancellation $\alpha + nb = b(n + \delta)$ remains visible; the integral identity has the advantage that it applies to other exponential-rational characteristics without rebuilding the calculation term by term. At $\alpha + b = 0$, the same conclusion follows by the finite resonant limit described after (2.7.54).

2.C Coefficient extraction from the generating functions

Let

$$G(x, y, t) = \sum_{i,j \geq 0} p_{i,j}(t) x^i y^j \quad (2.C.1)$$

be analytic in a neighbourhood of the origin. The bivariate Taylor formula gives

$$p_{i,j}(t) = [x^i y^j] G(x, y, t) = \frac{1}{i! j!} \left. \frac{\partial^{i+j} G}{\partial x^i \partial y^j} \right|_{x=y=0}. \quad (2.C.2)$$

For completeness, the coefficient is also represented by the bivariate Cauchy integral

$$p_{i,j}(t) = \frac{1}{(2\pi i)^2} \int_{|x|=r_x} \int_{|y|=r_y} \frac{G(x, y, t)}{x^{i+1} y^{j+1}} dy dx, \quad (2.C.3)$$

for circles inside the domain of analyticity. This form is useful for stable numerical extraction when derivatives of a special-function representation are unwieldy.

2.C.1 A multilinear power

Suppose

$$G(x, y, t) = \{c(t) + f(t)x + g(t)y\}^N. \quad (2.C.4)$$

For $a + b \leq N$, repeated differentiation gives

$$\frac{\partial^{a+b} G}{\partial x^a \partial y^b} = \frac{N!}{(N-a-b)!} f^a g^b (c + fx + gy)^{N-a-b}, \quad (2.C.5)$$

and the derivative is zero when $a + b > N$. Hence

$$p_{i,j}(t) = \frac{N!}{i! j! (N-i-j)!} f(t)^i g(t)^j c(t)^{N-i-j}, \quad i + j \leq N. \quad (2.C.6)$$

This is the trinomial law. In particular, the factor brought down by differentiation is the falling factorial $N!/(N-i-j)!$, not N^{i+j} . The distinction is essential for exact normalisation.

For transfer alone,

$$c = 0, \quad f = 1 - e^{-\alpha t}, \quad g = e^{-\alpha t}. \quad (2.C.7)$$

Only states with $i + j = N$ have positive probability, and

$$p_{i,j}(t) = \binom{N}{i} (1 - e^{-\alpha t})^i e^{-\alpha t j}, \quad i + j = N. \quad (2.C.8)$$

For transfer followed by death, take

$$\begin{aligned} c(t) &= 1 - \frac{\mu}{\mu - \alpha} e^{-\alpha t} + \frac{\alpha}{\mu - \alpha} e^{-\mu t}, \\ f(t) &= \frac{\alpha}{\mu - \alpha} (e^{-\alpha t} - e^{-\mu t}), \\ g(t) &= e^{-\alpha t}, \end{aligned} \quad (2.C.9)$$

with the continuous equal-rate limit from (2.7.17). Substitution into (2.C.6) gives the complete joint law. Since $c + f + g = 1$, summing over i and j recovers one by the multinomial theorem.

2.C.2 Transfer, birth and death

The full generating function is not a finite multilinear power in x , but its dependence on y remains linear before the power is taken. From (2.7.57),

$$G(x, y, t) = \{e^{-\alpha t}y + h(x, t)\}^N. \quad (2.C.10)$$

The binomial theorem gives

$$G(x, y, t) = \sum_{j=0}^N \binom{N}{j} e^{-\alpha t j} y^j h(x, t)^{N-j}, \quad (2.C.11)$$

and therefore

$$p_{i,j}(t) = \binom{N}{j} e^{-\alpha t j} [x^i] h(x, t)^{N-j}. \quad (2.C.12)$$

If

$$h(x, t) = \sum_{k \geq 0} h_k(t) x^k, \quad (2.C.13)$$

then the remaining coefficient is the finite convolution

$$[x^i] h(x, t)^M = \sum_{\substack{k_1 + \dots + k_M = i \\ k_r \geq 0}} \prod_{r=1}^M h_{k_r}(t). \quad (2.C.14)$$

Truncating h after degree i is exact for this calculation: coefficients of higher degree cannot contribute to $[x^i]$. This observation underlies `scripts/extract_coefficients.py`. As a check, summing over all interior states at fixed j means setting $x = 1$; since $h(1, t) = 1 - e^{-\alpha t}$,

$$\sum_{i=0}^{\infty} p_{i,j}(t) = \binom{N}{j} e^{-\alpha t j} (1 - e^{-\alpha t})^{N-j}, \quad (2.C.15)$$

which is the exterior binomial marginal in (2.7.48).

Bibliography

- [1] Theodore E. Harris. *The Theory of Branching Processes*, volume 119 of *Grundlehren der mathematischen Wissenschaften*. Springer-Verlag, Berlin, 1963.
- [2] Krishna B. Athreya and Peter E. Ney. *Branching Processes*, volume 196 of *Grundlehren der mathematischen Wissenschaften*. Springer-Verlag, Berlin, 1972.
- [3] Akiva M. Yaglom. Certain limit theorems of the theory of branching random processes. *Doklady Akademii Nauk SSSR*, 56:795–798, 1947.
- [4] Linda J. S. Allen. *An Introduction to Stochastic Processes with Applications to Biology*. CRC Press, Boca Raton, FL, 2 edition, 2010.
- [5] Gabriel Koenigs. Recherches sur les intégrales de certaines équations fonctionnelles. *Annales Scientifiques de l'École Normale Supérieure*, 1:3–41, 1884.
- [6] Paul-Georg Becker and Walter Bergweiler. Hypertranscendence of conjugacies in complex dynamics. *Mathematische Annalen*, 301(3):463–468, 1995.